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Documentation: Official Library Header Specification

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Maintained by: LDraw.org Library Administrator

Revision: 21-Feb-2025

Revision History:

29-Sep-2006: Added the "Unofficial_" types to !LDRAW_ORG meta

02-Oct-2006: Added "~Moved To" syntax

06-Sep-2007: Added Alias and Physical_Color as optional qualifiers

25-Dec-2015: Added defined description prefix character for programs that use parts.lst

13-Sep-2017: Added Flexible_Section optional qualifier

19-Aug-2019: Depreciated Physical_Color and made all parts the use it obsolete

22-Nov-2023: Added new !LICENSE lines

21-Feb-2025: Added !PREVIEW meta

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Purpose

LDraw Official Library header format suitable for the implementation of the [Contributor Agreement](#) (CA).

Scope

The header has to identify the LDraw.org username - the basis upon which CA affirmation is assessed - and if the part is covered by the CA or not. It has to conform to the new meta command syntax, where backwards compatibility allows this, as well as structure the history comment lines. No other meta-commands are allowed in the part header other than those specified in this document

Syntax

0 *PartDescription*

0 Name: *Filename.dat*

0 Author: *RealName* [*UserName*]

0 !LDRAW_ORG Part| Subpart| Primitive| 8_Primitive|

```

48_Primitive| Shortcut (optional qualifier(s))
ORIGINAL|UPDATE YYYY-RR
or
0 !LDRAW_ORG Unofficial_Part| Unofficial_Subpart|
Unofficial_Primitive| Unofficial_8_Primitive|
Unofficial_48_Primitive| Unofficial_Shortcut (optional
qualifier(s))
0 !LICENSE Licensed under CC BY 4.0 : see CReadme.txt
or
0 !LICENSE Licensed under CC BY 2.0 and CC BY 4.0 : see
CReadme.txt
or
0 !LICENSE Redistributable under CCAL version 2.0 : see
CReadme.txt
or
0 !LICENSE Not redistributable : see NonCReadme.txt

0 !HELP Optional free text description of file usage
0 !HELP Second row after user's line break to simulate paragraph

0 BFC ( CERTIFY ( CCW | CW ) | NOCERTIFY )

0 !CATEGORY Category name
0 !KEYWORDS words, more words,...,
0 !KEYWORDS words in second row, ..., final words

0 !CMDLINE LDraw run-time command(s)

0 !PREVIEW <colour> x y z a b c d e f g h i

0 !HISTORY YYYY-MM-DD [UserName] Free text description of change.
This can wrap to a
0 !HISTORY YYYY-MM-DD [UserName] second row with the same date if
necessary. However authors should lean toward writing longer
0 !HISTORY YYYY-MM-DD [UserName] single !HISTORY lines(and not
feel constrained to the historic 80-character limit on line length)
or
0 !HISTORY YYYY-MM-DD {RealName} Free text description of change

0 // Comments

1 <colour> x y z a b c d e f g h i <file>

```

Where:

PartDescription

General

This the descriptive name of the part. If the description contains dimensions, the numbers will be in decimal format (as opposed to fractions) and a leading space should be added to single digit numbers to aid in sorting. Decimals should be limited to 2 decimal places

Examples:

```

0 Brick  2 x  4
0 Brick  1 x 10 (Note the lack of a leading space on the 2 digit number)
0 Brick  3.1 x  3.1
0 Brick  3 x  3 x  0.34 (Note the 2 decimal places)

```

Needs Work

If the part is good enough for public use, but has some deficiencies that need to be addressed, the text " (Needs Work)" (without the quotation marks) can be added to the end of the description. If the description includes "(Needs Work)", a comment must be added to the file immediately after the header that explains the work that needs to be done

Special Prefix Characters

Some programs rely on the parts.lst file rather than the LDRAW_ORG meta-statement. Special prefix characters are added to the start of the part description to prevent the parts being included in the list by these programs:

- The descriptions of all subparts (i.e. line 4 = '0 !LDRAW_ORG Subpart') must start with '~'
- The description of all obsolete parts, maintained for backwards compatibility must start with '~'
- The use of '~' for non "s\" part files is at the author's discretion, to hide mouldings that are not released independently. This is one case where merely reading the LDRAW_ORG line is insufficient for tools to know how to treat the file.
- The descriptions of all aliases (i.e. line 4 = '0 !LDRAW_ORG Part Alias') must start with '='
- The descriptions of all third party parts must start with '|' or '~|' (as appropriate)
- The descriptions of all physical colour parts (i.e. line 4 = '0 !LDRAW_ORG Part Physical_Colour') should start with '_' (This usage is depreciated)

Name

Filename is the file name of the part including the folder (e.g. s/, 48/) if it is not directly in the parts or p folders.

Author

RealName is the author's real name.

UserName is the author's LDraw username. It is optional for those authors that had parts released prior to the establishment of the Parts Tracker and have not contributed since

!LDRAW_ORG

Official Parts

Part| Subpart| Primitive| 8_Primitive| 48_Primitive|

Shortcut are used in Official Library Parts

YYYY-RR is the LDraw update year and release within year.

Unofficial Parts

Unofficial_Part| Unofficial_Subpart| Unofficial_Primitive|

Unofficial_8_Primitive| Unofficial_48_Primitive|

Unofficial_Shortcut are used in unofficial parts on the Parts Tracker.

Optional Qualifiers

(optional qualifier(s)) are one of the following:

- **Alias**

The purpose of the Alias qualifier is to identify a part which is visually identical to another (often caused by LEGO using two different numbers, one for opaque and one for transparent versions of the same part (e.g. 32061), but more recently, there have also been new part numbers for well-established parts - such as brick 2x6

An alias file will typically, but not necessarily, contain only one sub-file reference (line type 1) entry

A file with an Alias type qualifier must not refer to a file that itself has an Alias qualifier.

- **Physical_Colour**

This qualifier is deprecated and all parts that used it have been made obsolete.

The purpose of the Physical_Colour qualifier was to identify a part which is encoded as a specific colour of a part

- **Flexible_Section**

The purpose of the Flexible_Section qualifier is to identify a file in the "parts" folder which represents a subsection of a flexible part. The placement of such files in the "parts" folder allows LDraw users and LDraw applications (such as LSynth) to utilise the subsections to create a flexed rendition of the parent part. Ideally LDraw editing applications should use this qualifier to identify flexible part subsections.

!LICENSE

CAreadme.txt or NonCAreadme.txt is the [human readable version of the CCAL](#) added to the parts library package. The Redistributable under CCAL version 2.0 and Not redistributable versions of this meta are deprecated and no longer used.

Optional Meta Commands

The following meta-commands are optional

- **!HELP** (Authors should constrain themselves to a 50-character limit to keep the text readable.)
- **!CMDLINE**
- **!PREVIEW <colour> x y z a b c d e f g h i**

This META defines how a part's preview image (e.g. the image generated for the Library part detail page or in a BOM) color, position, and rotation should be modified from the standard view. The standard view is defined as the part's bounding box centered on the origin and the camera positioned at 30 lat, 45 long looking at the origin.

<colour> x y z a b c d e f g h i

are the equivalent to the same definition as [type 1 line](#). Colours are limited to the colour numbers of the colours defined in the LDConfig.ldr

This META should not be included if the value is 16 0 0 0 1 0 0 0 1 0 0 0 1

- **BFC**

Note:

A 0 BFC CERTIFY CCW command is required for Official Library parts. For all other uses it is highly recommended to include this statement with a NOCERTIFY for non-BFC certified parts, but not required.

- !KEYWORDS
- !CATEGORY

Note:

This command required if the first word of the part description is not in the official category list: LDRAW.org [CATEGORY](#) and [KEYWORDS Language Extension](#). If !CATEGORY is set it must to be one of the listed categories.

Examples

```
0 Animal Scorpion
0 Name: 30169.dat
0 Author: Chris Dee [cwdee]
0 !LDRAW_ORG Part UPDATE 2006-01
0 !LICENSE Redistributable under CCAL version 2.0 : see
CAreadme.txt

0 !HELP Obviously there is no need for additional
0 !HELP information in this part. But it will give
0 !HELP you an idea of a primitive formatting.
0 !HELP
0 !HELP We should limit the length of
0 !HELP a line to 50 characters for the sake of readability.

0 BFC CERTIFY CW

0 !CATEGORY Animal
0 !KEYWORDS Sting, Poison, Adventurers, Egypt
0 !KEYWORDS Zodiac

0 !CMDLINE -c1

0 !HISTORY 2000-08-?? {Axel Poque} fixes to resolve L3P error
messages
0 !HISTORY 2000-08-?? {Manfred Moolhuysen} fixes to resolve L3P
error messages,
0 !HISTORY 2000-08-?? {Manfred Moolhuysen} fixed gap where body
meets head
0 !HISTORY 2002-04-25 [PTadmin] Official update 2002-02
0 !HISTORY 2004-05-18 [guyvivan] Made BFC compliant
0 !HISTORY 2004-05-18 [guyvivan] Used more primitives
0 !HISTORY 2006-??-?? [PTadmin] Official update 2006-01

0 // Tail
1 16 1 0 23 1 0 0 0 1 0 0 0 1 4-4cyli.dat

or

0 Boat Base 8 x 10
0 Name: 2622.dat
0 Author: Chris Alano
0 !LDRAW_ORG Part UPDATE 2000-02
0 !LICENSE Not redistributable : see NonCAreadme.txt
```

```
0 BFC NOCERTIFY
```

```
0 !KEYWORDS Pirates, Caribbean, Ship
```

```
2 24 100 24 80 80 24 20
```

Moved To Files

"~Moved to" files are created when part variants are created or the part real number for a "u" part is determined

Syntax

```
0 ~Moved to <new_number>
0 Name: <old_number>.dat
0 Author: [PTAdmin]
0 !LDRAW_ORG <Part, Subpart, etc as appropriate> (optional
qualifier(s)) UPDATE YYYY-RR
0 !LICENSE Redistributable under CCAL version 2.0 : see
CAreadme.txt

0 BFC CERTIFY CCW

0 !HISTORY <date> [PTAdmin] <any necessary comment(s)>

0 // <part_description>
1 16 0 0 0 1 0 0 0 1 0 0 0 1 <new_number>.dat
0
```

Example

```
0 ~Moved to 3842a
0 Name: 193a.dat
0 Author: [PTAdmin]
0 !LDRAW_ORG Part UPDATE 2006-01
0 !LICENSE Redistributable under CCAL version 2.0 : see
CAreadme.txt

0 BFC CERTIFY CCW

0 !HISTORY 2006-12-31 [PTAdmin] Official update 2006-01

0 // Minifig Helmet Classic with Thin Chin Guard
1 16 0 0 0 1 0 0 0 1 0 0 0 1 3842a.dat
0
```

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Parts Library Policies and FAQ

Documentation: LDraw.org Library Policies and FAQ

Policies for users of the Parts Tracker.

General

- Everyone at LDraw.org is a volunteer. This is our hobby and we have no paid staff.
- Always be respectful with all comments.

How to contact an admin

There are 2 ways to contact admin: the parts@ldraw.org or a post in Parts Tracker Discussion on the [LDraw.org Forum](#)

Gaining access as an author

- Register and confirm an account with the [LDraw.org Forum](#)
- Read the rest of this document and the [Contributor Agreement](#).
- Contact admin with your request. You will need to provide your true first and last name and your LDraw.org forum username. You can choose to use a pseudonym but this may have copyright implications (in general, nicknames or abbreviated forms of your true name are not considered pseudonyms).

The first time you submit a part to the Parts Tracker you will be asked to confirm your acceptance of the Contributor Agreement

Part sources

Do not submit parts that you did not author. We have a long standing informal agreement with LEGO to use shape data from LDD (now Studio and the Build app).

All other digital sources are strictly forbidden.

This includes data from licensed LEGO products (e.g. Traveler's Tales video games) or LDraw files distributed under other licenses, and other, non-LDraw, LEGO CAD products (e.g. MecaBricks). This is copyright theft and could result in legal action against you and/or LDraw.org. If you are unsure if you are able to use a source, ask the admins.

Do not submit parts by authors who have not agreed to the Contributor Agreement, even if those parts are licensed under the current library license.

Third Party Parts

Third party (i.e. non-LEGO produced) parts are allowed in the official library. Only those parts that are physical products and are produced with the intention of being used with official LEGO parts are allowed.

Clone brick products primary marketed to be used in lieu of LEGO parts are not allowed. When in doubt, ask the admins.

Since Third Party parts are often produced by Adult Fan of LEGO (AFOL) community members and in order to keep good community relations, the inclusion of a company's parts in the Official Library is strictly opt in. You must obtain permission from the company prior to submitting LDraw parts of their products to the Parts Tracker. This permission must be communicated to the library admins by a company's official representative. Since this rule was put into effect after the inclusion of some Third Party parts in the Official Library, those official parts will remain unless removal is requested by a company.

The following companies are authorized for the Parts Library

- Big Ben Bricks (<http://www.bigbenbricks.com/>)
- Brickstuff (<https://www.brickstuff.com/>)
- BuWizz (<https://buwizz.com>)
- Circuit Cubes (<https://circuitcubes.com/>)

The following companies are not authorized for the Parts Library

- BrickTrainDepot (<https://bricktraindepot.com/>)
- Fx Bricks (<https://shop.fxbricks.com/>)

Part Licensing

- All of an author's parts will use the same license. Per the Contributor Agreement, a part author can choose to release their parts with either the CC BY 4.0 license or the CC0 Public Domain declaration. CC0 is a permanent change. If an author chooses CC0 at any point, they cannot switch to CC BY 4.0.
- All CC BY 2 parts (official or unofficial) are free to be rewritten and replaced by any authorized parts author. This rewrite must be done **without** referencing the original file in any way. Rewritten parts will retain original !HISTORY with at least one additional !HISTORY line stating that the part has been rewritten and the name of the original author
- CC0 parts will not have their license changed

Editing parts

- For unofficial new parts (not fixes), you should always contact the author and ask their permission before changing their part. Respect and honor other peoples work. Give them a chance to look at and correct their mistakes. This allows authors, especially new authors, to learn and improve their skills. There should be a link to PM the author in the event feed if you wish to contact them.
- If an author is open to anyone making changes to their parts, especially if they plan on not working on a file any more they should put a note when they submit the part. A simple: "Anyone can make changes to this part" or "I've done all I can with it, anyone else is welcome to work on it" is all that is needed.
- Official part fixes are free to be edited by anyone. Likewise, if someone makes a relatively small change to an official part (like BFCing it), other people should feel free to make other fixes to it.
- New parts (and fixes) with a hold vote, which haven't been fixed within three months are open to anyone. The author has had the chance to respond to a hold vote, and if they haven't, at least they have had their chance to either correct the part or defend its correctness.
- All changes must be documented by a !HISTORY line if you are not the original parts author. Admin actions (i.e. header editing, renumbering) are exempt from this rule with the exception that all official part fixes must have at least one !HISTORY line documenting the nature of the change from the last time it was released.

Part Descriptions

Part Descriptions are written using Australian English terminology and spelling. Some common differences between American And Australian English include:

- Grey not Gray

- Tyre not Tire
- Colour not color
- Centre not Center

Note this is not an all inclusive list. If in doubt, ask a part reviewer or admin

Submitting parts of unknown number

If you don't know the official number for a part please ***DO NOT GUESS***. This causes additional unnecessary work for the admins, especially if there are subfiles involved. Used the uXXXX Part Number Reservation tab on the User Dashboard to request and track your assigned uXXXX numbers.

Renumbering/Moving Parts

If a part needs to be renumbered or moved, please ***DO NOT RESUBMIT WITH THE DESIRED NUMBER***. Doing this causes all the previous comments and actions on the part to be lost. Contact admin to have the part renumbered/moved.

Removing files from the Parts Tracker

- Removing parts is an admin only function. Please contact the admins to have this done.
- Hold the file with a comment to indicate to other users that deletion has been requested.
- Please don't resubmit the file with a modified description including the word 'deletable' - that probably will not get noticed by the admins.

Becoming a Part Reviewer

You first must establish yourself as a competent parts author. While there is no formal criteria for this, a general body of quality work, a solid familiarity with the standards, and regular community participation is a good start. If you feel that you are ready contact the admins to have your permissions upgraded.

How to do part reviews

Here's a non-comprehensive list of things in no particular order:

- Visually check the part very closely. Use a tool such as LDraw Part Editor. Check for missing lines, wrong condlines, warped quads, overlappings, gaps etc.
- Check the parts default orientation. Compare it to how similar parts are oriented and what orientation of the part would be useful.
- If the part is a fix to an already official part, ensure that the origin and orientation are unchanged.
- Check the part with [DATHeader](#). Make sure no error messages are produced.
- Check the Part for correct BFC.

Part Votes and Fast Track

In order to be included in a parts update, all unofficial files will require a minimum of two certify votes, one admin certify vote, and no hold votes

Administrators have the ability to cast a single "Fast Track" vote which will immediately qualify a part for release. Fast Track will not override the no hold vote requirement. Fast Track may only be used in the following conditions:

- An official part fix where geometry or origin is not affected
- Sticker or Minifig torso shortcuts where all component parts are official or in a Certified status
- Moved to parts where the origin of the moved part is not affected and the moved part is in a Certified status
- Obsolete files:
Files in the parts folder (types Part and Shortcut): geometry or origin is not affected and the replacement part is official or in a Certified status.

All other files: the part has no non-obsolete official or unofficial parent files. Typically these files are also emptied.

Part Update Schedule

As a general rule, a parts update will happen on the last weekend of each month. This schedule may flex based on admin availability or to allow a particular group of parts or fixes to finish the certification process.

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Documentation: LDraw.org Library Sticker Box Standard

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Maintained by: LDraw.org Library Administrators

Revision: 22-Jul-2025

Revision History:

22-Jul-2025: Original 2-Aug-2025: Reformatted for clarity. Specified orientation and naming for non-square/rectangular boxes.

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Sticker backing boxes

The backing box is defined as the sides and bottom surface of the sticker

For most stickers, the surface of the sticker is intended to be applied to an entire face of a single part. In this case, the size of the sticker should be that the edge of the sticker is 2 ldu from the edge of the part face. If actual, measured sticker size is +/- 1 ldu from this standard then the sticker should be modeled to the standard. Exceptions to this rule will be handled on a case by case basis and should be specifically addressed by the author upon submission.

Flat Sticker Backs

The backing boxes for older, sharp cornered stickers should be modeled with an appropriately scaled box5-12.

Circular and oval shaped stickers should use an appropriately scaled 4-4cylc3

Stickers with irregular shapes should use quads and triangles with conditional lines where appropriate.

Newer stickers have rounded instead of sharp cut corners. The radius of standard rounded corners will be 1.5 LDU.

Sticker Backs Files

A sticker back file is required for all square/rectangular with standard rounded corners. Sticker back files are optional for all other sticker shapes but encouraged if that shape and size is commonly reused.

All sticker back files shall be oriented such that the X width is greater with than the Z width (landscape format).

Sticker back files will be of the !LDRAW_ORG type of Subpart and will located in the parts/s/ directory.

Flat sticker back files shall use the follow naming convention:

stickerback<ZZZ>x<XXX><optional suffix a-z>.dat

Where:

<ZZZ> and <XXX> are a 3 digit numbers representing the Z width (for ZZZ) and X width (for XXX) of the sticker. The value of this number is LDU/20 rounded to the nearest tenth. A leading zeros will be added for a number less than 10. Example: if the Z width of the sticker is 143 ldu, ZZZ is 072.

Sticker back files without the optional suffix are reserved for the square/rectangluar sticker backs with standard rounded corners. Sticker backs of any other shape with the same dimensions shall use an optional suffix <a-z>.

Formed Sticker Backs

Orientation and origin of formed sticker backs shall follow the LDraw.org Official Parts Library Specifications for formed stickers.

Formed sticker back files should be created as reuse dictates.

Formed sticker backs shall use the follow naming convention:

<part number>bNN.dat

Where:

<part number> is the number of the part to which the sticker is formed NN is a 2 digit number with leading zeros starting at 01

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Maintained by: [LDraw.org Administration Team](#)

Revision: 17-Dec-2020

Revision History:

17-Dec-2020: Initial

22-Nov-2023: Update license requirements

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Introduction

The Official Model Repository (OMR) is a database of files in the LDraw File Format describing models that are released as sets by LEGO®.

For consistency between models and ease of indexing by software, a standard for the file headers, names, and hierarchy in the OMR is needed. This document will outline the extra requirements (in addition to those set forth in the [LDraw File Format](#) specification) for a model to be included in the OMR.

These requirements should be considered guidelines to be followed in most situations. Models that do not completely conform to this specification can still be included in the OMR at the discretion of the LDraw.org Administration Team

Base Requirements

All files in the model will conform to the current [LDraw File Format](#).

Base File Naming

Each model in the OMR will consist of several files that are packaged together into a single MPD file. For sets that include instructions for multiple models, each model will have its own MPD file.

For playsets or other sets where there are multiple models that are part of an integral whole, all of the submodels will be contained in one MPD. For example: the set 10267 - Gingerbread House has many different loose

models in the instructions. All of these models should be included in the same MPD and arranged in a master scene.

Each MPD for the set will be named in the following manner:

`<Set Number>[-<Qualifier Number>] - <Set Name>[ - <Sub Model Name>]`

Where:

`<Set Number>`: The number assigned on the container of the set.

`-<Qualifier Number>`: Is a sequential number, starting with 1. The qualifier of "-1" is optional and gets added only if there is more than one set that could be assigned the same `<Set Number>`. The first set using a given number would be understood to never contain the qualifier however numbering should start with the oldest set and some investigation should be done in existing set databases. All subsequent numbers are required.

Example:

6901 – Mobile Lab.mpd (Produced in 1980)

6901-2 – Space Plane.mpd (Produced in 1998)

`<Set Name>`: The name of the set printed on the container in English.

`<Sub Model Name>`: This is optional in most cases. This is required for alternate models that are detailed in instructions (e.g. the Creator theme). In this case the naming is left to the discretion of the author but should be descriptive of the model contained in the MPD.

META commands

All META commands are allowed in the model file but not specifically required except as specified for the header. If included, any META commands used should enable any instructions generated to be as close to the official instructions as possible.

Mirrored geometry

Models with bilaterally symmetric sections can often be modeled much more easily by modeling one half of the symmetric section and then including it in the model twice, with one instance mirrored around the symmetry line. Unfortunately, this mirroring produces an incorrect parts list, and also produces a mirrored LEGO logo on the model's studs when rendered with certain software, so using mirroring is strongly discouraged.

MPD File Structure

The MPD file will conform to the [MPD File Specification](#).

Regular files and subfiles

The filename of each file in the MPD will have the following structure:

`<Set Number>[-<Qualifier Number>] - <Individual filename>`

Where:

`<Set Number>` and `-<Qualifier Number>` are described above and should match the parent MPD file name.

<Individual filename> is up to the discretion of the author with the following guidance:

- A logical naming scheme is highly desired.
- Each individual model in the set (e.g. a vehicle or minifig) shall have its own separate file inside the MPD.
- Minifig file name should have the name of the character, if known.

Unofficial Parts

Unofficial parts are allowed to be used but must be included in the MPD as referenced subfiles. The filename of the unofficial part is subject to the same naming rules as outlined above except that <Individual filename> will be the part number of the unofficial part as it appears in the referring type 1 line

Example

33956.dat would be renamed to 3345 – 33956.dat or 3345–1 – 33956.dat.

It is highly encouraged that any parts created for use in an OMR file be submitted to the LDraw.org Parts Tracker.

Unavailable Parts

If a part is unavailable either officially or on the LDraw.org Parts Tracker, a suitable substitution may be made. If the unavailable part is a patterned part with an unpatterned version available, use the unpatterned version.

If a substitution or omission is made a comment stating such is desired, but not required. Reference the step and page number of the instructions for clarity, if needed

Examples:

0 // The next piece should have the Star Wars Hatch pattern, Bricklink Number XXXX

0 // Bionicle piece X should go here per step Y on page Z

File Headers

Each individual model file in the MPD that does not represent an unofficial part must have the following standard header format.

Each individual model file in the MPD that represents an unofficial part must use the standard header format of unofficial parts on the parts tracker. Its 0 FILE entry will reflect the modified MPD version of the part's filename, but the rest of the header will reflect the original filename.

Standard Header:

```
0 FILE <Filename>.ldr
0 <Individual filename>
0 Name: <Filename>.ldr
0 Author: <Author Name> [Username]
0 !LDraw_ORG Model -OR- 0 !LDraw_ORG Unofficial_Model
0 !LICENSE Licensed under CC BY 4.0 : see CReadme.txt

0 !THEME Theme name
0 !KEYWORDS words, more words, ...,
```

0 !KEYWORDS words in second row, ..., final words

0 !HISTORY YYYY-MM-DD [Username] Free text description of change.
This can wrap to a
0 !HISTORY YYYY-MM-DD [Username] second row with the same date if
necessary. However authors should lean toward writing longer
0 !HISTORY YYYY-MM-DD [Username] single !HISTORY lines (and not
feel constrained to the historic 80-character limit on line
length).

Where:

<Filename> and <Individual filename> are as described above

!THEME is the LEGO branded theme of the set. Where possible try to match
this with other inventory website.

All other meta commands are described in [Official Library Header
Specification](#)

The commands are !THEME, !KEYWORDS, and !HISTORY are optional

Example:

```
0 FILE 4896 - Roadster Main.ldr
0 Roadster Main
0 Name: 4896 - Roadster Main.ldr
0 Author: Joe Smith [jsmith]
0 !LDRAW_ORG Unofficial_Model
0 !LICENSE Licensed under CC BY 4.0 : see CReadme.txt

0 !THEME Creator
0 !KEYWORDS car, convertible

0 !HISTORY 2011-08-01 [jsmith] Initial creation
```

OMR Example

```
0 FILE 6712 - Main.ldr
0 Main
0 Name: 6712 - Main.ldr
0 Author: Willy Tschager [Holly-Wood]
0 !LDRAW_ORG Model
0 !LICENSE Licensed under CC BY 4.0 : see CReadme.txt

1 0 0 0 0 1 0 0 0 1 0 0 0 1 6712 - Fireplace.ldr
1 0 -140 -49 -32 0.5 0 -0.866 0 1 0 0.866 0 0.5 6712 - Horse.ldr
1 0 -140.866 -129 -31.5 0.5 0 -0.866 0 1 0 0.866 0 0.5 6712 -
Sheriff.ldr
1 0 83 -99 64 1 0 0 0 1 0 0 0 1 6712 - Bandit.ldr
1 6 81 3 76 1 0 0 0 1 0 0 0 1 30139.dat
1 2 -51 8 86 1 0 0 0 1 0 0 0 1 6064.dat
1 7 80 0 -14 0.707 0 0.707 0 1 0 -0.707 0 0.707 3069bp03.dat

0 FILE 6712 - Horse.ldr
0 Horse
0 Name: 6712 - Horse.ldr
0 Author: Willy Tschager [Holly-Wood]
0 !LDRAW_ORG Model
```

0 !LICENSE Licensed under CC BY 4.0 : see CReadme.txt

1 0 0 0 0 1 0 0 0 1 0 0 0 1 4493c01.dat
1 6 0 -8 0 1 0 0 0 1 0 0 0 1 4491b.dat
1 7 -31 -44 27 1 0 0 0 0 -1 0 1 0 30141.dat

0 FILE 6712 - Fireplace.ldr
0 Fireplace
0 Name: 6712 - Fireplace.ldr
0 Author: Willy Tschager [Holly-Wood]
0 !LDRAW_ORG Model
0 !LICENSE Licensed under CC BY 4.0 : see CReadme.txt

1 7 0 0 0 1 0 0 0 1 0 0 0 1 3031.dat
0 STEP
1 0 -30 -8 10 0 0 -1 0 1 0 1 0 0 2555.dat
1 0 -30 -8 -10 0 0 -1 0 1 0 1 0 0 2555.dat
0 STEP
1 0 0 -8 -10 1 0 0 0 1 0 0 0 1 2412b.dat
1 0 0 -8 10 1 0 0 0 1 0 0 0 1 2412b.dat
0 STEP
1 38 -24 -14 -10 0 0 1 0 -1 0 1 0 0 6126a.dat
1 38 -24 -14 10 0 0 1 0 -1 0 1 0 0 6126a.dat
1 0 0 -28 -2 -0.966 0 0.259 0 1 0 -0.259 0 -0.966 4528.dat

0 FILE 6712 - Sheriff.ldr
0 Sheriff
0 Name: 6712 - Sheriff.ldr
0 Author: Willy Tschager [Holly-Wood]
0 !LDRAW_ORG Model
0 !LICENSE Licensed under CC BY 4.0 : see CReadme.txt

1 7 0 -24 0 1 0 0 0 1 0 0 0 1 3629pw2.dat
1 14 0 -24 0 1 0 0 0 1 0 0 0 1 3626bp35.dat
1 7 0 0 0 1 0 0 0 1 0 0 0 1 973pw4.dat
1 0 0 32 0 1 0 0 0 1 0 0 0 1 3815.dat
1 7 -15.552 9 0 0.985 -0.128 -0.112 0.17 0.743 0.646 0 -0.656
0.755 3818.dat
1 7 15.552 9 0 0.985 0.139 0.098 -0.17 0.807 0.565 0 -0.574 0.819
3819.dat
1 0 -21.93 15.393 -20.098 0.995 -0.066 0.069 0.066 0.997 0.007
-0.07 -0.002 0.997 3820.dat
1 0 22.28 17.422 -19.215 0.985 0.167 -0.03 -0.17 0.97 -0.171 0
0.174 0.985 3820.dat
1 0 0 44 0 1 0 0 0 1 0 0 0 1 3816.dat
1 0 0 44 0 1 0 0 0 1 0 0 0 1 3817.dat
1 7 -22.656 13.187 -30.096 0.998 -0.052 0.033 0.058 0.975 -0.213
-0.021 0.214 0.976 30132.dat

0 FILE 6712 - Bandit.ldr
0 Bandit
0 Name: 6712 - Bandit.ldr
0 Author: Willy Tschager [Holly-Wood]
0 !LDRAW_ORG Model
0 !LICENSE Licensed under CC BY 4.0 : see CReadme.txt

1 6 0 -27 0 1 0 0 0 1 0 0 0 1 3629.dat

```
1 14 0 -27 0 1 0 0 0 1 0 0 0 1 3626bpw1.dat
1 0 0 0 0 -1 0 0 0 1 0 0 0 -1 30133.dat
1 0 0 0 0 1 0 0 0 1 0 0 0 1 973pw5.dat
1 0 0 32 0 1 0 0 0 1 0 0 0 1 3815.dat
1 0 -15.552 9 0 0.985 -0.128 -0.112 0.17 0.743 0.646 0 -0.656
0.755 3818.dat
1 0 15.552 9 0 0.985 0.139 0.098 -0.17 0.807 0.565 0 -0.574 0.819
3819.dat
1 0 -21.93 15.393 -20.098 0.135 0.99 0.012 -0.988 0.136 -0.069
-0.07 -0.002 0.997 3820.dat
1 0 22.28 17.422 -19.215 0.985 0.167 -0.03 -0.17 0.97 -0.171 0
0.174 0.985 3820.dat
1 7 22.08 16.263 -29.297 0.985 0.155 -0.069 -0.17 0.899 -0.401 0
0.407 0.913 30132.dat
1 0 0 44 0 1 0 0 0 1 0 0 0 1 3816.dat
1 0 0 44 0 1 0 0 0 1 0 0 0 1 3817.dat
1 2 -18 13 -47 0.259 -0.118 0.959 0 0.993 0.122 -0.966 -0.032
0.257 3069bpw2.dat
```

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Rules and procedures for the Official Model Repository



Documentation: Rules and Procedures for the Official Model Repository

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Maintained by: LDraw.org Administration Team

Revision: 22-Nov-2023

Revision History:

18-Jun-2020: Initial

22-Nov-2023: Update license requirements

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Guidelines for the Operation of the Official Model Repository

- The OMR located at <http://library.ldraw.org/omr> is linked from the homepage via Downloads->Official Model Repository (OMR).
- The OMR is named "official" because it contains only LDraw replicas of sets (and their submodels) released by the LEGO Group. MOCs or alterations official models will not be considered.
- The models will conform to the [Official Model Repository \(OMR\) Specification](#). It is at the discretion of the LDraw.org Administration Team or their representatives to "OMRize" submitted models if they do not conform.
- The OMR will only consider new models that contain the `0 !LICENSE` Licensed under CC BY 4.0 : see `CAREadme.txt` statement ensuring that it is within the guidelines of the [LDraw.org Contributor Agreement](#). All embedded parts will retain the original, unaltered part license line. Legacy and model updates will use `0 !LICENSE` Licensed under CC BY 2.0 and CC BY 4.0 : see `CAREadme.txt` unless it is absolutely certain that the original author has agreed to release the model under the CC BY 4.0 license
- The OMR will host the actual LDraw file. Links to external model files are not possible.
- All file uploads, updates, etc. are handled by the the LDraw.org Administration Team. There is currently no mechanism for a model author to manage their own files.
- The model author will upload the model to the [LDraw.org Forums](#). Once any discussion regarding the model has finished, the LDraw.org Administration Team will copy the uploaded file to the OMR at their earliest convenience.

- Updates of files made after the first official release will contain a !HISTORY line, which states the date, author and a short description of the change. Updates will be posted as a reply to the post of the original version.

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Documentation: Colour Definition Reference

Colour Definitions

The LDraw colour definitions are contained in `LDConfig.ldr` file. This file is distributed with the LDraw.org Official Library.

The current version of LDConfig.ldr:

<https://library.ldraw.org/library/official/LDConfig.ldr>

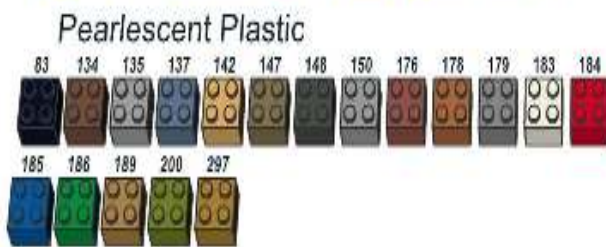
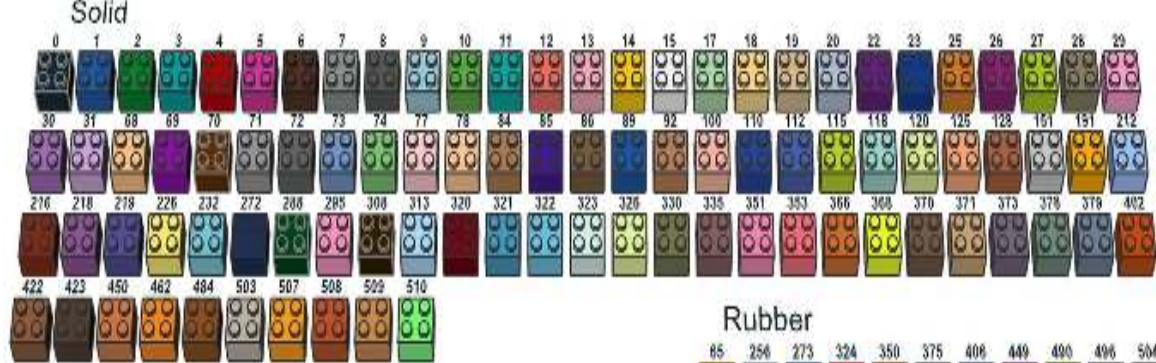
The current version of LDCfgalt.ldr:

<https://library.ldraw.org/library/official/LDCfgalt.ldr>

Detailed information about scope, syntax and the definitions of parameters of how colours are defined is contained in the [Colour Definition \(!COLOUR\) Language Extension](#).

Visual LDConfig

The picture shows many of the colours defined in the colour definition file `ldconfig.ldr` as rendered by LDView



Colour Chart

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Black	CODE 0	VALUE	#1B2A34	EDGE	#808080
0 !COLOUR	Blue	CODE 1	VALUE	#1E5AA8	EDGE	#333333
0 !COLOUR	Green	CODE 2	VALUE	#00852B	EDGE	#333333
0 !COLOUR	Dark_Turquoise	CODE 3	VALUE	#069D9F	EDGE	#333333
0 !COLOUR	Red	CODE 4	VALUE	#B40000	EDGE	#333333
0 !COLOUR	Dark_Pink	CODE 5	VALUE	#D3359D	EDGE	#333333
0 !COLOUR	Brown	CODE 6	VALUE	#543324	EDGE	#1E1E1E
0 !COLOUR	Light_Grey	CODE 7	VALUE	#8A928D	EDGE	#333333
0 !COLOUR	Dark_Grey	CODE 8	VALUE	#545955	EDGE	#333333
0 !COLOUR	Light_Blue	CODE 9	VALUE	#97CBD9	EDGE	#333333
0 !COLOUR	Bright_Green	CODE 10	VALUE	#58AB41	EDGE	#333333

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Light_Turquoise	CODE 11	VALUE	#00AAA4	EDGE	#333333
0 !COLOUR	Salmon	CODE 12	VALUE	#F06D61	EDGE	#333333
0 !COLOUR	Pink	CODE 13	VALUE	#F6A9BB	EDGE	#333333
0 !COLOUR	Yellow	CODE 14	VALUE	#FAC80A	EDGE	#333333
0 !COLOUR	White	CODE 15	VALUE	#F4F4F4	EDGE	#333333
0 !COLOUR	Light_Green	CODE 17	VALUE	#ADD9A8	EDGE	#333333
0 !COLOUR	Light_Yellow	CODE 18	VALUE	#FFD67F	EDGE	#333333
0 !COLOUR	Tan	CODE 19	VALUE	#D7BA8C	EDGE	#333333
0 !COLOUR	Light_Violet	CODE 20	VALUE	#AFBED6	EDGE	#333333
0 !COLOUR	Purple	CODE 22	VALUE	#671F81	EDGE	#333333
0 !COLOUR	Dark_Blue_Violet	CODE 23	VALUE	#0E3E9A	EDGE	#333333
0 !COLOUR	Orange	CODE 25	VALUE	#D67923	EDGE	#333333
0 !COLOUR	Magenta	CODE 26	VALUE	#901F76	EDGE	#333333
0 !COLOUR	Lime	CODE 27	VALUE	#A5CA18	EDGE	#333333
0 !COLOUR	Dark_Tan	CODE 28	VALUE	#897D62	EDGE	#333333
0 !COLOUR	Bright_Pink	CODE 29	VALUE	#FF9ECD	EDGE	#333333
0 !COLOUR	Medium_Lavender	CODE 30	VALUE	#A06EB9	EDGE	#333333
0 !COLOUR	Lavender	CODE 31	VALUE	#CDA4DE	EDGE	#333333
0 !COLOUR	Very_Light_Orange	CODE 68	VALUE	#FDC383	EDGE	#333333
0 !COLOUR	Bright_Reddish_Lilac	CODE 69	VALUE	#8A12A8	EDGE	#333333
0 !COLOUR	Reddish_Brown	CODE 70	VALUE	#5F3109	EDGE	#808080
0 !COLOUR	Light_Bluish_Grey	CODE 71	VALUE	#969696	EDGE	#333333
0 !COLOUR	Dark_Bluish_Grey	CODE 72	VALUE	#646464	EDGE	#333333
0 !COLOUR	Medium_Blue	CODE 73	VALUE	#7396C8	EDGE	#333333
0 !COLOUR	Medium_Green	CODE 74	VALUE	#7FC475	EDGE	#333333
0 !COLOUR	Light_Pink	CODE 77	VALUE	#FECCCF	EDGE	#333333
0 !COLOUR	Light_Nougat	CODE 78	VALUE	#FFC995	EDGE	#333333
0 !COLOUR	Medium_Nougat	CODE 84	VALUE	#AA7D55	EDGE	#333333
0 !COLOUR	Medium_Lilac	CODE 85	VALUE	#441A91	EDGE	#333333
0 !COLOUR	Light_Brown	CODE 86	VALUE	#7B5D41	EDGE	#333333
0 !COLOUR	Blue_Violet	CODE 89	VALUE	#1C58A7	EDGE	#333333

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Nougat	CODE 92	VALUE	#BB805A	EDGE	#333333
0 !COLOUR	Light_Salmon	CODE 100	VALUE	#F9B7A5	EDGE	#333333
0 !COLOUR	Violet	CODE 110	VALUE	#26469A	EDGE	#333333
0 !COLOUR	Medium_Violet	CODE 112	VALUE	#4861AC	EDGE	#333333
0 !COLOUR	Medium_Lime	CODE 115	VALUE	#B7D425	EDGE	#333333
0 !COLOUR	Aqua	CODE 118	VALUE	#9CD6CC	EDGE	#333333
0 !COLOUR	Light_Lime	CODE 120	VALUE	#DEEA92	EDGE	#333333
0 !COLOUR	Light_Orange	CODE 121	VALUE	#F89A39	EDGE	#333333
0 !COLOUR	Dark_Salmon	CODE 123	VALUE	#EE5434	EDGE	#333333
0 !COLOUR	Spud_Orange	CODE 125	VALUE	#F9A777	EDGE	#333333
0 !COLOUR	Dark_Nougat	CODE 128	VALUE	#AD6140	EDGE	#333333
0 !COLOUR	Very_Light_Bluish_Grey	CODE 151	VALUE	#C8C8C8	EDGE	#333333
0 !COLOUR	Dark_Yellow	CODE 180	VALUE	#DD982E	EDGE	#333333
0 !COLOUR	Bright_Light_Orange	CODE 191	VALUE	#FCAC00	EDGE	#333333
0 !COLOUR	Bright_Light_Blue	CODE 212	VALUE	#9DC3F7	EDGE	#333333
0 !COLOUR	Medium_Blue_Violet	CODE 213	VALUE	#476FB6	EDGE	#333333
0 !COLOUR	Rust	CODE 216	VALUE	#872B17	EDGE	#333333
0 !COLOUR	Reddish_Lilac	CODE 218	VALUE	#8E5597	EDGE	#333333
0 !COLOUR	Lilac	CODE 219	VALUE	#564E9D	EDGE	#333333
0 !COLOUR	Light_Lilac	CODE 220	VALUE	#9195CA	EDGE	#333333
0 !COLOUR	Warm_Yellowish_Orange	CODE 225	VALUE	#FAA964	EDGE	#333333
0 !COLOUR	Bright_Light_Yellow	CODE 226	VALUE	#FFEC6C	EDGE	#333333
0 !COLOUR	Sky_Blue	CODE 232	VALUE	#77C9D8	EDGE	#333333
0 !COLOUR	Dark_Blue	CODE 272	VALUE	#19325A	EDGE	#333333
0 !COLOUR	Dark_Green	CODE 288	VALUE	#00451A	EDGE	#808080
0 !COLOUR	Flamingo_Pink	CODE 295	VALUE	#FF94C2	EDGE	#333333
0 !COLOUR	Dark_Brown	CODE 308	VALUE	#352100	EDGE	#808080
0 !COLOUR	Maersk_Blue	CODE 313	VALUE	#ABD9FF	EDGE	#333333
0 !COLOUR	Dark_Red	CODE 320	VALUE	#720012	EDGE	#333333
0 !COLOUR	Dark_Azure	CODE 321	VALUE	#469BC3	EDGE	#333333
0 !COLOUR	Medium_Azure	CODE 322	VALUE	#68C3E2	EDGE	#333333

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Light_Aqua	CODE 323	VALUE	#D3F2EA	EDGE	#333333
0 !COLOUR	Yellowish_Green	CODE 326	VALUE	#E2F99A	EDGE	#333333
0 !COLOUR	Olive_Green	CODE 330	VALUE	#77774E	EDGE	#333333
0 !COLOUR	Sand_Red	CODE 335	VALUE	#88605E	EDGE	#333333
0 !COLOUR	Medium_Dark_Pink	CODE 351	VALUE	#F785B1	EDGE	#333333
0 !COLOUR	Coral	CODE 353	VALUE	#FF6D77	EDGE	#333333
0 !COLOUR	Earth_Orange	CODE 366	VALUE	#D86D2C	EDGE	#333333
0 !COLOUR	Neon_Yellow	CODE 368	VALUE	#EDFF21	EDGE	#333333
0 !COLOUR	Medium_Brown	CODE 370	VALUE	#755945	EDGE	#333333
0 !COLOUR	Medium_Tan	CODE 371	VALUE	#CCA373	EDGE	#333333
0 !COLOUR	Sand_Purple	CODE 373	VALUE	#75657D	EDGE	#333333
0 !COLOUR	Sand_Green	CODE 378	VALUE	#708E7C	EDGE	#333333
0 !COLOUR	Sand_Blue	CODE 379	VALUE	#70819A	EDGE	#333333
0 !COLOUR	Reddish_Orange	CODE 402	VALUE	#CA4C0B	EDGE	#333333
0 !COLOUR	Sienna_Brown	CODE 422	VALUE	#915C3C	EDGE	#333333
0 !COLOUR	Umber_Brown	CODE 423	VALUE	#543F33	EDGE	#333333
0 !COLOUR	Ochre_Yellow	CODE 424	VALUE	#DD9E47	EDGE	#333333
0 !COLOUR	Warm_Pink	CODE 430	VALUE	#F79B9B	EDGE	#333333
0 !COLOUR	Bright_Blue_Violet	CODE 431	VALUE	#665EA7	EDGE	#333333
0 !COLOUR	Fabuland_Brown	CODE 450	VALUE	#D27744	EDGE	#333333
0 !COLOUR	Medium_Orange	CODE 462	VALUE	#F58624	EDGE	#333333
0 !COLOUR	Dark_Orange	CODE 484	VALUE	#91501C	EDGE	#333333
0 !COLOUR	Very_Light_Grey	CODE 503	VALUE	#BCB4A5	EDGE	#333333
0 !COLOUR	Fabuland_Red	CODE 508	VALUE	#C65127	EDGE	#333333
0 !COLOUR	Fabuland_Orange	CODE 509	VALUE	#CF8A47	EDGE	#333333
0 !COLOUR	Fabuland_Lime	CODE 510	VALUE	#ACE35A	EDGE	#333333
0 !COLOUR	Lemon	CODE 10015	VALUE	#FFF230	EDGE	#333333
0 !COLOUR	Rose_Pink	CODE 10017	VALUE	#FF9494	EDGE	#333333
0 !COLOUR	Yellowish_Dark_Pink	CODE 10022	VALUE	#D05098	EDGE	#333333
0 !COLOUR	Trans_Black_IR_Lens	CODE 32	VALUE	#000000	EDGE	#333333

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Trans_Dark_Blue	CODE 33	VALUE	#0020A0	EDGE	#000B38
0 !COLOUR	Trans_Green	CODE 34	VALUE	#237841	EDGE	#174F2B
0 !COLOUR	Trans_Bright_Green	CODE 35	VALUE	#56E646	EDGE	#27AF18
0 !COLOUR	Trans_Red	CODE 36	VALUE	#C91A09	EDGE	#660D00
0 !COLOUR	Trans_Dark_Pink	CODE 37	VALUE	#DF6695	EDGE	#B9275F
0 !COLOUR	Trans_Neon_Orange	CODE 38	VALUE	#FF800D	EDGE	#A85100
0 !COLOUR	Trans_Very_Light_Blue	CODE 39	VALUE	#C1DFF0	EDGE	#6FB4D0
0 !COLOUR	Trans_Brown	CODE 40	VALUE	#635F52	EDGE	#2A2823
0 !COLOUR	Trans_Medium_Blue	CODE 41	VALUE	#559AB7	EDGE	#326276
0 !COLOUR	Trans_Neon_Green	CODE 42	VALUE	#C0FF00	EDGE	#739900
0 !COLOUR	Trans_Light_Blue	CODE 43	VALUE	#AEE9EF	EDGE	#59D1DE
0 !COLOUR	Trans_Bright_Reddish_Lilac	CODE 44	VALUE	#96709F	EDGE	#5F4368
0 !COLOUR	Trans_Pink	CODE 45	VALUE	#FC97AC	EDGE	#F9345E
0 !COLOUR	Trans_Yellow	CODE 46	VALUE	#F5CD2F	EDGE	#B49208
0 !COLOUR	Trans_Clear	CODE 47	VALUE	#FCFCFC	EDGE	#C9C9C9
0 !COLOUR	Trans_Purple	CODE 52	VALUE	#A5A5CB	EDGE	#6464A0
0 !COLOUR	Trans_Neon_Yellow	CODE 54	VALUE	#DAB000	EDGE	#755E00
0 !COLOUR	Trans_Orange	CODE 57	VALUE	#F08F1C	EDGE	#9E5C00
0 !COLOUR	Trans_Neon_Red	CODE 158	VALUE	#F18EBB	EDGE	#E63384
0 !COLOUR	Trans_Bright_Light_Green	CODE 227	VALUE	#B5D96C	EDGE	#86B22E

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Trans_Bright_Light_Orange	CODE 231	VALUE	#FCB76D	EDGE	#FA860A
0 !COLOUR	Trans_Fire_Yellow	CODE 234	VALUE	#FBE890	EDGE	#F7D22B
0 !COLOUR	Trans_Reddish_Lilac	CODE 284	VALUE	#C281A5	EDGE	#944771
0 !COLOUR	Trans_Light_Green	CODE 285	VALUE	#7DC291	EDGE	#469550
0 !COLOUR	Trans_Light_Blue_Violet	CODE 293	VALUE	#6BABE4	EDGE	#247BC6
0 !COLOUR	Trans_Black	CODE 10375	VALUE	#212121	EDGE	#545454
0 !COLOUR	Trans_White	CODE 11015	VALUE	#F4F4F4	EDGE	#C1C1C1
0 !COLOUR	Trans_Tan	CODE 11019	VALUE	#D7BA8C	EDGE	#BC8D4A
0 !COLOUR	Chrome_Antique_Brass	CODE 60	VALUE	#645A4C	EDGE	#665B41
0 !COLOUR	Chrome_Blue	CODE 61	VALUE	#6C96BF	EDGE	#3D638A
0 !COLOUR	Chrome_Green	CODE 62	VALUE	#3CB371	EDGE	#226741
0 !COLOUR	Chrome_Pink	CODE 63	VALUE	#AA4D8E	EDGE	#632C53
0 !COLOUR	Chrome_Black	CODE 64	VALUE	#1B2A34	EDGE	#3D5F76
0 !COLOUR	Chrome_Gold	CODE 334	VALUE	#DFC176	EDGE	#C29821
0 !COLOUR	Chrome_Silver	CODE 383	VALUE	#CECECE	EDGE	#9C9C9C
0 !COLOUR	Pearl_Black	CODE 83	VALUE	#0A1327	EDGE	#333333
0 !COLOUR	Copper	CODE 134	VALUE	#764D3B	EDGE	#333333
0 !COLOUR	Pearl_Light_Grey	CODE 135	VALUE	#A0A0A0	EDGE	#333333
0 !COLOUR	Metallic_Blue	CODE 137	VALUE	#5B7590	EDGE	#333333
0 !COLOUR	Pearl_Light_Gold	CODE 142	VALUE	#DEAC66	EDGE	#333333
0 !COLOUR	Pearl_Dark_Gold	CODE 147	VALUE	#83724F	EDGE	#333333
0 !COLOUR	Pearl_Dark_Grey	CODE 148	VALUE	#484D48	EDGE	#333333
0 !COLOUR	Pearl_Very_Light_Grey	CODE 150	VALUE	#989B99	EDGE	#333333
0 !COLOUR	Pearl_Red	CODE 176	VALUE	#945148	EDGE	#333333
0 !COLOUR	Pearl_Yellow	CODE 178	VALUE	#AB673A	EDGE	#333333
0 !COLOUR	Pearl_Silver	CODE 179	VALUE	#898788	EDGE	#333333

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Pearl_White	CODE 183	VALUE	#F6F2DF	EDGE	#333333
0 !COLOUR	Metallic_Bright_Red	CODE 184	VALUE	#D60026	EDGE	#333333
0 !COLOUR	Metallic_Bright_Blue	CODE 185	VALUE	#0059A3	EDGE	#333333
0 !COLOUR	Metallic_Dark_Green	CODE 186	VALUE	#008E3C	EDGE	#333333
0 !COLOUR	Pearl_Brown	CODE 187	VALUE	#57392C	EDGE	#333333
0 !COLOUR	Reddish_Gold	CODE 189	VALUE	#AC8247	EDGE	#333333
0 !COLOUR	Lemon_Metallic	CODE 200	VALUE	#708224	EDGE	#333333
0 !COLOUR	Cool_Silver	CODE 296	VALUE	#ADADAD	EDGE	#333333
0 !COLOUR	Pearl_Gold	CODE 297	VALUE	#AA7F2E	EDGE	#333333
0 !COLOUR	Flat_Silver	CODE 315	VALUE	#8C8C8C	EDGE	#333333
0 !COLOUR	Titanium_Metallic	CODE 316	VALUE	#3E3C39	EDGE	#333333
0 !COLOUR	Conductive_Black	CODE 342	VALUE	#3E3C39	EDGE	#333333
0 !COLOUR	Reddish_Copper	CODE 346	VALUE	#803D07	EDGE	#333333
0 !COLOUR	Metallic_Silver	CODE 80	VALUE	#767676	EDGE	#333333
0 !COLOUR	Metallic_Green	CODE 81	VALUE	#C2C06F	EDGE	#333333
0 !COLOUR	Metallic_Gold	CODE 82	VALUE	#DBAC34	EDGE	#333333
0 !COLOUR	Metallic_Dark_Grey	CODE 87	VALUE	#3E3C39	EDGE	#333333
0 !COLOUR	Metallic_Copper	CODE 300	VALUE	#C27F53	EDGE	#333333
0 !COLOUR	Metallic_Light_Blue	CODE 10045	VALUE	#97CBD9	EDGE	#333333
0 !COLOUR	Metallic_Pink	CODE 10046	VALUE	#AD659A	EDGE	#333333
0 !COLOUR	Metallic_Light_Pink	CODE 10049	VALUE	#FECCCF	EDGE	#333333
0 !COLOUR	Fluorescent_Red_Ink	CODE 332	VALUE	#D06D4F	EDGE	#333333
0 !COLOUR	Fluorescent_Green_Ink	CODE 333	VALUE	#DBEA7A	EDGE	#333333
0 !COLOUR	Glow_In_Dark_Opaque	CODE 21	VALUE	#E0FFB0	EDGE	#B8FF4D
0 !COLOUR	Milky_White	CODE 79	VALUE	#EEEEEE	EDGE	#BABAB
0 !COLOUR	Glow_In_Dark_Trans	CODE 294	VALUE	#BDC6AD	EDGE	#8D9D7
0 !COLOUR	Glow_In_Dark_White	CODE 329	VALUE	#F5F3D7	EDGE	#E0DA8
0 !COLOUR	Glitter_Trans_Dark_Pink	CODE 114	VALUE	#DF6695	EDGE	#B9275F

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Glitter_Trans_Clear	CODE 117	VALUE	#EEEEEE	EDGE	#BABAB
0 !COLOUR	Glitter_Trans_Purple	CODE 129	VALUE	#640061	EDGE	#000000
0 !COLOUR	Glitter_Trans_Light_Blue	CODE 302	VALUE	#AEE9EF	EDGE	#59D1D1
0 !COLOUR	Glitter_Trans_Neon_Green	CODE 339	VALUE	#C0FF00	EDGE	#739900
0 !COLOUR	Glitter_Trans_Orange	CODE 341	VALUE	#F08F1C	EDGE	#9E5C00
0 !COLOUR	Glitter_Trans_Bright_Green	CODE 10351	VALUE	#56E646	EDGE	#27AF18

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Opal_Trans_Clear	CODE 360	VALUE	#FCFCFC	EDGE	#C9C9C9
0 !COLOUR	Opal_Trans_Light_Blue	CODE 362	VALUE	#AEE9EF	EDGE	#59D1D1
0 !COLOUR	Opal_Trans_Black	CODE 363	VALUE	#635F52	EDGE	#2A2823
0 !COLOUR	Opal_Trans_Dark_Pink	CODE 364	VALUE	#DF6695	EDGE	#B9275F
0 !COLOUR	Opal_Trans_Purple	CODE 365	VALUE	#671F81	EDGE	#441456
0 !COLOUR	Opal_Trans_Green	CODE 367	VALUE	#237841	EDGE	#174F2B

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Opal_Trans_Yellow	CODE 376	VALUE	#F5CD2F	EDGE	#B49208
0 !COLOUR	Opal_Trans_Dark_Blue	CODE 10366	VALUE	#0020A0	EDGE	#000B38
0 !COLOUR	Speckle_Black_Copper	CODE 75	VALUE	#000000	EDGE	#AB6038
0 !COLOUR	Speckle_Dark_Bluish_Grey_Silver	CODE 76	VALUE	#635F61	EDGE	#898788
0 !COLOUR	Speckle_Black_Silver	CODE 132	VALUE	#000000	EDGE	#898788
0 !COLOUR	Speckle_Black_Gold	CODE 133	VALUE	#000000	EDGE	#DBAC38

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Modulex_Clear	CODE 30000	VALUE	#FCFCFC	EDGE	#C9C9C9
0 !COLOUR	Modulex_White	CODE 30001	VALUE	#F4F4F4	EDGE	#333333
0 !COLOUR	Modulex_Light_Grey	CODE 30002	VALUE	#8A928D	EDGE	#333333
0 !COLOUR	Modulex_Black	CODE 30003	VALUE	#34434D	EDGE	#333333
0 !COLOUR	Modulex_Terracotta	CODE 30004	VALUE	#D27744	EDGE	#333333
0 !COLOUR	Modulex_Buff	CODE 30005	VALUE	#D7BA8C	EDGE	#333333
0 !COLOUR	Modulex_Ochre_Yellow	CODE 30006	VALUE	#FED557	EDGE	#333333
0 !COLOUR	Modulex_Olive_Green	CODE 30007	VALUE	#7C9051	EDGE	#333333
0 !COLOUR	Modulex_Teal_Blue	CODE 30008	VALUE	#467083	EDGE	#333333
0 !COLOUR	Modulex_Brown	CODE 30009	VALUE	#907450	EDGE	#333333
0 !COLOUR	Modulex_Strong_Red	CODE 30010	VALUE	#B40000	EDGE	#333333
0 !COLOUR	Modulex_Pastel_Blue	CODE 30011	VALUE	#ABD9FF	EDGE	#333333
0 !COLOUR	Modulex_Orange	CODE 30012	VALUE	#D86D2C	EDGE	#333333
0 !COLOUR	Modulex_Red	CODE 30013	VALUE	#C65127	EDGE	#333333
0 !COLOUR	Modulex_Pastel_Green	CODE 30014	VALUE	#78FC78	EDGE	#333333
0 !COLOUR	Modulex_Lemon	CODE 30015	VALUE	#FFF230	EDGE	#333333
0 !COLOUR	Modulex_Pink	CODE 30016	VALUE	#F785B1	EDGE	#333333
0 !COLOUR	Modulex_Light_Bluish_Grey	CODE 30032	VALUE	#AFB5C7	EDGE	#333333
0 !COLOUR	Modulex_Pink_Red	CODE 30033	VALUE	#F45C40	EDGE	#333333
0 !COLOUR	Modulex_Aqua_Green	CODE 30034	VALUE	#27867E	EDGE	#333333
0 !COLOUR	Modulex_Light_Yellow	CODE 30035	VALUE	#FFE371	EDGE	#333333
0 !COLOUR	Modulex_Violet	CODE 30037	VALUE	#BD7D85	EDGE	#333333
0 !COLOUR	Modulex_Medium_Blue	CODE 30038	VALUE	#61AFFF	EDGE	#333333
0 !COLOUR	Modulex_Light_Orange	CODE 30039	VALUE	#F7AD63	EDGE	#333333
0 !COLOUR	Modulex_Charcoal_Grey	CODE 30040	VALUE	#595D60	EDGE	#333333
0 !COLOUR	Modulex_Dark_Brown	CODE 30054	VALUE	#330000	EDGE	#808080
0 !COLOUR	Modulex_Foil_White	CODE 30080	VALUE	#F4F4F4	EDGE	#333333
0 !COLOUR	Modulex_Foil_Black	CODE 30081	VALUE	#34434D	EDGE	#333333
0 !COLOUR	Modulex_Foil_Dark_Grey	CODE 30082	VALUE	#595D60	EDGE	#333333

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Modulex_Foil_Light_Grey	CODE 30083	VALUE	#9C9C9C	EDGE	#333333
0 !COLOUR	Modulex_Foil_Dark_Green	CODE 30084	VALUE	#006400	EDGE	#808080
0 !COLOUR	Modulex_Foil_Light_Green	CODE 30085	VALUE	#7DB538	EDGE	#333333
0 !COLOUR	Modulex_Foil_Dark_Blue	CODE 30086	VALUE	#0057A6	EDGE	#333333
0 !COLOUR	Modulex_Foil_Light_Blue	CODE 30087	VALUE	#68AECE	EDGE	#333333
0 !COLOUR	Modulex_Foil_Violet	CODE 30088	VALUE	#4B0082	EDGE	#333333
0 !COLOUR	Modulex_Foil_Dark_Red	CODE 30089	VALUE	#8B0000	EDGE	#333333
0 !COLOUR	Modulex_Foil_Yellow	CODE 30090	VALUE	#FED557	EDGE	#333333
0 !COLOUR	Modulex_Foil_Orange	CODE 30091	VALUE	#F7AD63	EDGE	#333333
0 !COLOUR	Modulex_Dark_Grey	CODE 31000	VALUE	#6B5A5A	EDGE	#333333
0 !COLOUR	Modulex_Blue	CODE 31001	VALUE	#0057A6	EDGE	#333333
0 !COLOUR	Modulex_Very_Light_Grey	CODE 31002	VALUE	#D9D5CC	EDGE	#333333
0 !COLOUR	Rubber_Yellow	CODE 65	VALUE	#FAC80A	EDGE	#333333
0 !COLOUR	Rubber_Black	CODE 256	VALUE	#1B2A34	EDGE	#808080
0 !COLOUR	Rubber_Blue	CODE 273	VALUE	#1E5AA8	EDGE	#333333
0 !COLOUR	Rubber_Red	CODE 324	VALUE	#B40000	EDGE	#333333
0 !COLOUR	Rubber_Orange	CODE 350	VALUE	#D67923	EDGE	#333333
0 !COLOUR	Rubber_Light_Grey	CODE 375	VALUE	#8A928D	EDGE	#333333
0 !COLOUR	Rubber_Dark_Blue	CODE 406	VALUE	#19325A	EDGE	#333333
0 !COLOUR	Rubber_Purple	CODE 449	VALUE	#671F81	EDGE	#333333
0 !COLOUR	Rubber_Lime	CODE 490	VALUE	#A5CA18	EDGE	#333333
0 !COLOUR	Rubber_Light_Bluish_Grey	CODE 496	VALUE	#969696	EDGE	#333333
0 !COLOUR	Rubber_Flat_Silver	CODE 504	VALUE	#898788	EDGE	#333333
0 !COLOUR	Rubber_White	CODE 511	VALUE	#F4F4F4	EDGE	#333333
0 !COLOUR	Rubber_Green	CODE 10002	VALUE	#00852B	EDGE	#333333
0 !COLOUR	Rubber_Dark_Turquoise	CODE 10003	VALUE	#069D9F	EDGE	#333333
0 !COLOUR	Rubber_Dark_Pink	CODE 10005	VALUE	#D3359D	EDGE	#333333
0 !COLOUR	Rubber_Dark_Grey	CODE 10008	VALUE	#545955	EDGE	#333333
0 !COLOUR	Rubber_Bright_Green	CODE 10010	VALUE	#58AB41	EDGE	#333333
0 !COLOUR	Rubber_Tan	CODE 10019	VALUE	#D7BA8C	EDGE	#333333
0 !COLOUR	Rubber_Magenta	CODE 10026	VALUE	#901F76	EDGE	#333333

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Rubber_Dark_Tan	CODE 10028	VALUE	#897D62	EDGE	#333333
0 !COLOUR	Rubber_Bright_Pink	CODE 10029	VALUE	#FF9ECD	EDGE	#333333
0 !COLOUR	Rubber_Medium_Lavender	CODE 10030	VALUE	#A06EB9	EDGE	#333333
0 !COLOUR	Rubber_Lavender	CODE 10031	VALUE	#CDA4DE	EDGE	#333333
0 !COLOUR	Rubber_Reddish_Brown	CODE 10070	VALUE	#5F3109	EDGE	#808080
0 !COLOUR	Rubber_Dark_Bluish_Grey	CODE 10072	VALUE	#646464	EDGE	#333333
0 !COLOUR	Rubber_Medium_Blue	CODE 10073	VALUE	#7396C8	EDGE	#333333
0 !COLOUR	Rubber_Light_Nougat	CODE 10078	VALUE	#FFC995	EDGE	#333333
0 !COLOUR	Rubber_Medium_Nougat	CODE 10084	VALUE	#AA7D55	EDGE	#333333
0 !COLOUR	Rubber_Medium_Lilac	CODE 10085	VALUE	#441A91	EDGE	#333333
0 !COLOUR	Rubber_Nougat	CODE 10092	VALUE	#BB805A	EDGE	#333333
0 !COLOUR	Rubber_Pearl_Light_Grey	CODE 10135	VALUE	#A0A0A0	EDGE	#333333
0 !COLOUR	Rubber_Metallic_Blue	CODE 10137	VALUE	#5B7590	EDGE	#333333
0 !COLOUR	Rubber_Pearl_Dark_Gold	CODE 10147	VALUE	#83724F	EDGE	#333333
0 !COLOUR	Rubber_Pearl_Dark_Grey	CODE 10148	VALUE	#484D48	EDGE	#333333
0 !COLOUR	Rubber_Pearl_Very_Light_Grey	CODE 10150	VALUE	#989B99	EDGE	#333333
0 !COLOUR	Rubber_Very_Light_Bluish_Grey	CODE 10151	VALUE	#C8C8C8	EDGE	#333333
0 !COLOUR	Rubber_Reddish_Gold	CODE 10189	VALUE	#AC8247	EDGE	#333333
0 !COLOUR	Rubber_Bright_Light_Orange	CODE 10191	VALUE	#FCAC00	EDGE	#333333
0 !COLOUR	Rubber_Lemon_Metallic	CODE 10200	VALUE	#708224	EDGE	#333333
0 !COLOUR	Rubber_Bright_Light_Blue	CODE 10212	VALUE	#9DC3F7	EDGE	#333333
0 !COLOUR	Rubber_Bright_Light_Yellow	CODE 10226	VALUE	#FFEC6C	EDGE	#333333
0 !COLOUR	Rubber_Dark_Green	CODE 10288	VALUE	#00451A	EDGE	#808080
0 !COLOUR	Rubber_Pearl_Gold	CODE 10297	VALUE	#AA7F2E	EDGE	#333333
0 !COLOUR	Rubber_Dark_Brown	CODE 10308	VALUE	#352100	EDGE	#808080
0 !COLOUR	Rubber_Dark_Red	CODE 10320	VALUE	#720012	EDGE	#333333
0 !COLOUR	Rubber_Dark_Azure	CODE 10321	VALUE	#469BC3	EDGE	#333333
0 !COLOUR	Rubber_Medium_Azure	CODE 10322	VALUE	#68C3E2	EDGE	#333333
0 !COLOUR	Rubber_Light_Aqua	CODE 10323	VALUE	#D3F2EA	EDGE	#333333
0 !COLOUR	Rubber_Yellowish_Green	CODE 10326	VALUE	#E2F99A	EDGE	#333333
0 !COLOUR	Rubber_Olive_Green	CODE 10330	VALUE	#77774E	EDGE	#333333

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Rubber_Sand_Green	CODE 10378	VALUE	#708E7C	EDGE	#333333
0 !COLOUR	Rubber_Dark_Orange	CODE 10484	VALUE	#91501C	EDGE	#333333
0 !COLOUR	Rubber_Trans_Yellow	CODE 66	VALUE	#F5CD2F	EDGE	#B49208
0 !COLOUR	Rubber_Trans_Clear	CODE 67	VALUE	#FCFCFC	EDGE	#C9C9C9
0 !COLOUR	Rubber_Trans_Bright_Green	CODE 10035	VALUE	#56E646	EDGE	#27AF18
0 !COLOUR	Rubber_Trans_Red	CODE 10036	VALUE	#C91A09	EDGE	#660D0D
0 !COLOUR	Rubber_Trans_Light_Blue	CODE 10043	VALUE	#AEE9EF	EDGE	#59D1D1
0 !COLOUR	Rubber_Trans_Orange	CODE 10057	VALUE	#F08F1C	EDGE	#9E5C0A
0 !COLOUR	Rubber_Milky_White	CODE 10079	VALUE	#EEEEEE	EDGE	#BABABA
0 !COLOUR	Canvas_Black	CODE 20000	VALUE	#1B2A34	EDGE	#808080
0 !COLOUR	Canvas_Blue	CODE 20001	VALUE	#1E5AA8	EDGE	#333333
0 !COLOUR	Canvas_Green	CODE 20002	VALUE	#00852B	EDGE	#333333
0 !COLOUR	Canvas_Red	CODE 20004	VALUE	#B40000	EDGE	#333333
0 !COLOUR	Canvas_Brown	CODE 20006	VALUE	#543324	EDGE	#1E1E1E
0 !COLOUR	Canvas_Light_Grey	CODE 20007	VALUE	#8A928D	EDGE	#333333
0 !COLOUR	Canvas_Bright_Green	CODE 20010	VALUE	#58AB41	EDGE	#333333
0 !COLOUR	Canvas_Yellow	CODE 20014	VALUE	#FAC80A	EDGE	#333333

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Canvas_White	CODE 20015	VALUE	#F4F4F4	EDGE	#333333
0 !COLOUR	Canvas_Tan	CODE 20019	VALUE	#D7BA8C	EDGE	#333333
0 !COLOUR	Canvas_Purple	CODE 20022	VALUE	#671F81	EDGE	#333333
0 !COLOUR	Canvas_Dark_Tan	CODE 20028	VALUE	#897D62	EDGE	#333333
0 !COLOUR	Canvas_Reddish_Brown	CODE 20070	VALUE	#5F3109	EDGE	#808080
0 !COLOUR	Canvas_Light_Bluish_Grey	CODE 20071	VALUE	#969696	EDGE	#333333
0 !COLOUR	Canvas_Dark_Bluish_Grey	CODE 20072	VALUE	#646464	EDGE	#333333
0 !COLOUR	Canvas_Light_Brown	CODE 20086	VALUE	#7B5D41	EDGE	#333333
0 !COLOUR	Canvas_Dark_Green	CODE 20288	VALUE	#00451A	EDGE	#808080
0 !COLOUR	Canvas_Dark_Red	CODE 20320	VALUE	#720012	EDGE	#333333
0 !COLOUR	Canvas_Sand_Red	CODE 20335	VALUE	#88605E	EDGE	#333333
0 !COLOUR	Canvas_Cream	CODE 20500	VALUE	#F8F3E4	EDGE	#333333
0 !COLOUR	Obsolete_Light_Orange_Brown	CODE 507	VALUE	#FA9C1C	EDGE	#333333
0 !COLOUR	Main_Colour	CODE 16	VALUE	#FFFF80	EDGE	#333333
0 !COLOUR	Edge_Colour	CODE 24	VALUE	#7F7F7F	EDGE	#333333
0 !COLOUR	Magnet	CODE 493	VALUE	#656761	EDGE	#333333
0 !COLOUR	Electric_Contact_Alloy	CODE 494	VALUE	#D0D0D0	EDGE	#333333

0 !COLOUR	Name	CODE x	VALUE	v	EDGE	e
0 !COLOUR	Electric_Contact_Copper	CODE 495	VALUE	#AE7A59	EDGE	#333333
0 !COLOUR	Trans_Sticker	CODE 10047	VALUE	#FFFFFF	EDGE	#FFFFFF

Colour Inclusion Considerations

The main considerations on how LDraw colours are defined include:

- Use of LEGO RGB values
- Use of LEGO numbering where possible
- Use of Bricklink names
- All colours currently listed at Bricklink
- Lists the matched LEGO name
- Colours are grouped in categories

There is also an alternative file distributed with the LDraw.org Official Library that contains different edgeline colours: `LDCfgAlt.ldr`. It is primarily intended for rendering and those who prefer their LDraw models to look more realistic in an editor. To use this alternative file, you will need to rename `LDConfig.ldr` to something like `LDConfig_BlackEdges.ldr` and rename `LDConfig_Alt.ldr` to `LDConfig.ldr`. Your chosen LDraw editor will then load the alternative version at the next start-up. Please don't use `LDConfig_Alt.ldr` to make instructions.

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Documentation: Common Error Check Messages

Use this list as a guide when reviewing parts. When you are writing parts, it is best to resolve **all** error messages.

ITEM	TYPE	CAUSE	VERDICT
1	ERROR	Embedded POV Code found.	HOLD
2	ERROR	'0 WRITE' found. -	HOLD
3	ERROR	'0 BFC CERTIFY INVERTNEXT' found.	HOLD
4	ERROR	'0 ROTATION' found.	HOLD
5	ERROR	'0 COLOR' found.	HOLD
6	ERROR	Incorrect use of only '0' for comments. Should be '0 //' inst	HOLD
7	ERROR	Not all used colors in LDConfig.ldr.	HOLD
8	ERROR	Matrix all zero found.	HOLD
9	ERROR	Identical vertices found.	HOLD
10	ERROR	Colinear vertices found.	HOLD
11	ERROR	Bad Vertex Sequence found.	HOLD
12	ERROR	Concave quads found.	HOLD
13	ERROR	Coplanarity.	HOLD
14	ERROR	Incorrect color for linetype. Must be 24 for edges and conds, while 1, 3, 4 cannot get 24	HOLD
15	ERROR	Loop in reference found.	HOLD
16	ERROR	Double lines found.	HOLD
19	WARNING	Maybe leading spaces in part description.	WARNING

ITEM	TYPE	CAUSE	VERDICT
21	ERROR	Some keywords are also in part description.	HOLD
22	ERROR	Filename does not match file title.	HOLD
23	ERROR	Author real name is not set	HOLD
24	ERROR	Author user name is not set	HOLD
25	ERROR	BFC is not set.	HOLD
26	ERROR	License is not set.	HOLD
27	ERROR	Part type is not set.	HOLD
28	ERROR	Primitives needs to have CCW winding.	HOLD
29	ERROR	Use of !KEYWORDS is not allowed for this part type.	WARNING
30	ERROR	Part description contains Tab character.	HOLD
31	ERROR	HELP entry length	WARNING
32	ERROR	Word 'new' or 'old' is used in part description	WARNING
34	ERROR	Entry for !CATEGORY	HOLD
35	ERROR	Use of !CATEGORY is not allowed at for this part type	WARNING
36	ERROR	High-res primitive has to start with '48\.'	HOLD
37	ERROR	Description for primitives should not start with '_ ' or '~'. Unless it is a Moveto or Obsoloted	HOLD
38	ERROR	Filename for parts, shortcuts and primitives should not start with '48\, '8\ or 's\.'	HOLD
39	ERROR	Filename for subparts has to start with 's\.'	HOLD
40	ERROR	Description for subparts has to start with '~'.	HOLD
41	ERROR	Description for subparts should not start with '_ ' or '= '	HOLD
45	ERROR	Extension has to be .dat	HOLD
47	ERROR	Lines do not end with <CR><LF>.	HOLD
48	ERROR	Author !HISTORY entry has no brackets or curly braces	HOLD
49	ERROR	Not scalable primitive is scaled.	HOLD

ITEM	TYPE	CAUSE	VERDICT
50	ERROR	Primitive is scaled in illegal direction	HOLD
51	ERROR	First line after BFC INVERTNEXT isn't linetype 1	HOLD
52	ERROR	Problem with the RGB value.	HOLD
53	ERROR	'~Moved to' file used.	HOLD
54	ERROR	Wrong BFC command found.	HOLD
56	ERROR	'Minifig Accessory' found in part description	HOLD
57	ERROR	'Figure Accessory' found in part description	HOLD
58	WARNING	'INVERTNEXT' used although not necessary.	WARNING
59	WARNING	T-Junction detected.	WARNING
60	ERROR	Wrong brackets around username used. Only [] is allowed	WARNING
61	ERROR	Description for alias parts should start with '='	HOLD
65	ERROR	Only linetype 0 and 1 are allowed in Alias parts.	HOLD
67	ERROR	Only color 16 is allowed in Alias parts.	HOLD
73	WARNING	Mirrored studs detected.	WARNING
74	WARNING	Flat subfile scaled in flat direction	WARNING
75	ERROR	Optional line with same vertices than line found.	HOLD
76	ERROR	Leading or trailing zeros found.	HOLD
77	ERROR	Overlapping triangle found.	WARNING
78	ERROR	Lines with wrong number of arguments found.	HOLD
79	WARNING	Filename ending does not match declared LDRAW_ORG file type.	WARNING
80	ERROR	Moved To number with extension. ~Moved to 3004.dat	HOLD
81	WARNING	Filename ending does not match declared LDRAW_ORG part type qualifier.	WARNING

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Maintained by: LDraw.org Steering Committee**Revision:** 2024-06-06**Revision History:**

2022-02-23: Added new Contributor's Agreements authorizing use of CC BY 4.0 license.

2024-06-06: Added an option for CC0 license

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Legacy CA

Version: 4.2 (23-Oct-04)

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You must also include the statement "I accept the LDraw.org Contributor Agreement with regards to all past and future contributions I make to LDraw.org". A list of the current authors can be found at

<http://www.ldraw.org/cgi-bin/tracker/ptuserlist.cgi>

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Documentation: Legal Info

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- [LDraw All-In-One-Installer Agreement](#)
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LDraw.org, July 2007

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LDraw.org, October 2008

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Should any person wish to use LDraw models for building instructions or other rendered images for sale or otherwise the LDraw SteerCo would appreciate that the items include due credit to reward the efforts of all those who have contributed to the library. Furthermore, where possible, the SteerCo would also appreciate a donation be made to help defray server costs. This is, however, not by any means a requirement.

Paying tribute to the fact that LDraw.org is a completely volunteer organization the SteerCo prefers workmanship contribution, such as part authoring, parts reviewing, writing programs or tutorials over financial return.

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Documentation: LDraw File Format Specification

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Maintained by: LDraw.org Standards Board

Revision: 1.0.2, 29-Apr-2012

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Purpose

Describing LEGO® bricks and models in a virtual environment is a challenging prospect. In 1995 James Jessiman developed just such a system with his LDraw program and file format. Since then the LDraw file format has grown to be the standard by which most create LEGO® models on the computer.

The aim of this document is to compile and consolidate James's original file specification with selected additions made since.

LDraw Files

Text Based

All LDraw files are text based.

Text Encoding

LDraw files must be created in the UTF-8 encoding of the Unicode character set. The byte order mark (BOM) must not appear in LDraw files. LDraw software may choose to reject files containing the BOM.

Note:

Previous versions of the LDraw.org File Format did not specify a text encoding for LDraw files. Software not using UTF-8 has been widely deployed. Parsers are recommended to confirm that a file is valid UTF-8, and if not, to attempt either (a) opening the file using the Microsoft Windows code page most appropriate for the current language, or (b) open the file using Windows Code Page 1252. These fallback recommendations are encouraged for all software platforms, since they follow the behavior of the overwhelming majority of non-Unicode LDraw editors.

- [CATEGORY and KEYWORDS Statements](#)
- [MPD Statements](#)
- [BFC Statements](#)
- [Unofficial META Commands](#)
- [Adding New META Commands](#)

File Name

The requirement for LDraw files to conform to the DOS 8.3 filename convention has been removed. Filenames should not exceed 255 characters (including the extension). Special characters, such as &, #, |, and ?, should be avoided as they may also cause cross-platform issues and create problems when used in URLs. It is noted that long filenames and filenames with non-DOS permitted characters may not work with (older) software that does not fully support this specification.

Extension

All LDraw files carry the LDR (default), DAT or MPD extension.

Basic parsing / file content

An LDraw file consists of one command per line. There is no line length restriction. Each command consists of optional leading whitespace followed by whitespace-delimited tokens. Some commands also have trailing arbitrary data which may itself include internal whitespace; such data is not tokenized, but treated as single unit according to the command.

Whitespace is defined as one or more spaces (#32), tabs (#9), or combination thereof.

Lines may also be empty or consist only of whitespace. Such lines have no effect.

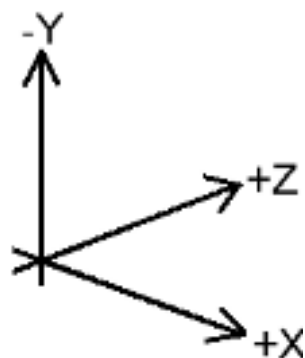
If a line is non-empty, the first token must be an integer from the list of valid Line Type numbers. This number dictates further parsing of tokens for that line. The parsing rules per linetype follow below.

Line Termination

All lines in the file must use the standard DOS/Windows line termination of <CR><LF> (carriage return/line feed). The file is permitted (but not required) to end with a <CR><LF>. It is recommended that all LDraw-compliant programs also be capable of reading files with the standard Unix line termination of <LF> (line feed).

The LDraw Co-ordinate System

LDraw uses a right-handed co-ordinate system where -Y is "up".



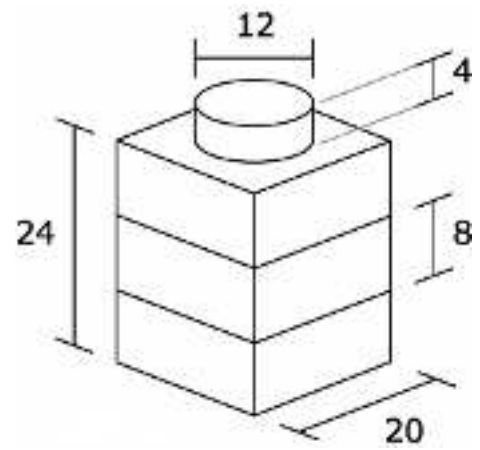
The LDraw Unit (LDU)

LDraw parts are measured in LDraw Units (LDU)

- 1 brick width/depth = 20 LDU
- 1 brick height = 24 LDU
- 1 plate height = 8 LDU
- 1 stud diameter = 12 LDU
- 1 stud height = 4 LDU

Real World Approximations

- 1 LDU = 1/64 in
- 1 LDU = 0.4 mm



Note:

These real world approximations are just that: approximations.

While they can be used for determining the LDraw equivalent of a portion of a physical part (for example, the curve of a surface), key points on such portions should be verified by reference to interactions with real parts. For example, it is usually possible to build a framework of bricks and plates to "hug" the curve and act as reference points. When using these dimensions to determine a pattern they should only be used for small areas/detail. Large areas should be measured in multiples of plate/brick dimensions with any "left-over" being converted with the above approximations.

Line Types

The line type of a line is the first number on the line. The line types are:

- 0: Comment or META Command
- 1: Sub-file reference
- 2: Line
- 3: Triangle
- 4: Quadrilateral
- 5: Optional Line

If the line type of the command is invalid, the line is ignored.

Line Type 0

Line type 0 has two uses. One use is a comment the other is as a META command.

If the first line of a file is a line type 0 the remainder of the line is considered the file title (see [Library Header specification](#)). This overrides any META commands on that line.

A comment line is formatted:

```
0 // <comment>
```

or

```
0 <comment>
```

Where:

<comment> is any string

The form `0 // <comment>` is preferred as the `//` marker clearly indicates that the line is a comment, thereby permitting parsers to stop processing the line.

The form `0 <comment>` is deprecated.

META Commands

A META command is a statement used to tell an LDraw compatible program to do something. There are currently many official META commands and even more [unofficial META commands](#). In a META command, a keyword follows the line type in the line. The keyword must be in all caps. The generic META line format is:

`0 !<META command> <additional parameters>`

Where:

- `!` is used to positively identify this as a META command. (Note: A few official meta commands do not start with a `!` in order to preserve backwards compatibility, however, all new official META commands must start with a `!` and it is strongly recommended that new unofficial meta-commands also start with a `!`)
- `<META command>` is any string in all caps
- `<additional parameters>` is any string. Note that if a META command does not require any additional parameter, none should be given.

Line Type 1

Line type 1 is a sub-file reference. The generic format is:

`1 <colour> x y z a b c d e f g h i <file>`

Where:

- `<colour>` is a number representing the colour of the part. See the [Colours section](#) for allowable colour numbers.
- `x y z` is the x y z coordinate of the part
- `a b c d e f g h i` is a top left 3x3 matrix of a standard 4x4 homogeneous transformation matrix. This represents the rotation and scaling of the part. The entire 4x4 3D transformation matrix would then take either of the following forms:

$$\begin{array}{ccc} / a & d & g \ 0 \backslash & / a & b & c & x \backslash \\ | b & e & h \ 0 | & | d & e & f & y | \\ | c & f & i \ 0 | & | g & h & i & z | \\ \backslash x & y & z & 1 / & \backslash 0 & 0 & 0 & 1 / \end{array}$$

The above two forms are essentially equivalent, but note the location of the transformation portion (x, y, z) relative to the other terms.

- Formally, the transformed point (u', v', w') can be calculated from point (u, v, w) as follows:

$$\begin{aligned} u' &= a*u + b*v + c*w + x \\ v' &= d*u + e*v + f*w + y \\ w' &= g*u + h*v + i*w + z \end{aligned}$$

- `<file>` is the filename of the sub-file referenced and must be a [valid LDraw filename](#). Any leading and/or trailing whitespace must be ignored.

Line type 1 should never use the colour 24.

The complement colour is neither unique nor symmetric. The complement of the complement of colour X is not necessarily colour X. Using colour 24 for line type 1 will therefore lead to unexpected/undefined results.

Sub-files can be located in the LDRAW\PARTS sub-directory, the LDRAW\P sub-directory, the LDRAW\MODELS sub-directory, the current file's directory, a path relative to one of these directories, or a full path may be specified. Sub-parts are typically stored in the LDRAW\PARTS\S sub-directory and so are referenced as s\subpart.dat, while hi-res primitives are stored in the LDRAW\P\48 sub-directory and so referenced as 48\hires.dat

While there is no specified limit on how deep sub-files may be nested, there are probably practical limitations imposed by individual software programs.

There are many on-line references about transformation matrices, one such reference is [3D Transformations – Part 1 Matrices](#)

Line Type 2

Line type 2 is a line drawn between two points. The generic format is:

```
2 <colour> x1 y1 z1 x2 y2 z2
```

Where:

- <colour> is a number representing the colour of the part, typically this is 24 - the edge colour. See the [Colours section](#) for allowable colour numbers.
- x1 y1 z1 is the coordinate of the first point
- x2 y2 z2 is the coordinate of the second point

Line type 2 (and also 5) is typically used to edge parts. When used in this manner colour 24 must be used for the line. It should be remembered that not all renderers display line types 2 and 5

Line Type 3

Line type 3 is a filled triangle drawn between three points. The generic format is:

```
3 <colour> x1 y1 z1 x2 y2 z2 x3 y3 z3
```

Where:

- <colour> is a number representing the colour of the part. See the [Colours section](#) for allowable colour numbers.
- x1 y1 z1 is the coordinate of the first point
- x2 y2 z2 is the coordinate of the second point
- x3 y3 z3 is the coordinate of the third point

See also the comments about polygons at the end of the Line Type 4 section.

Line Type 4

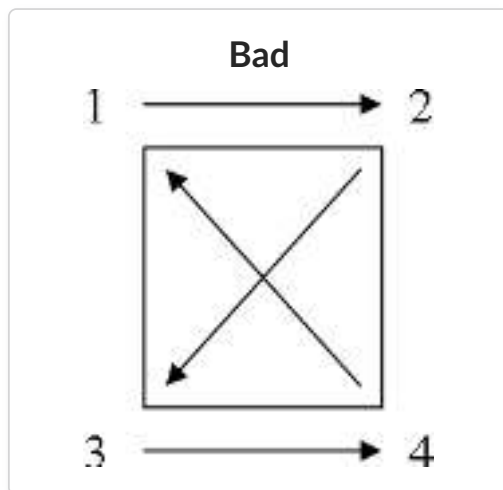
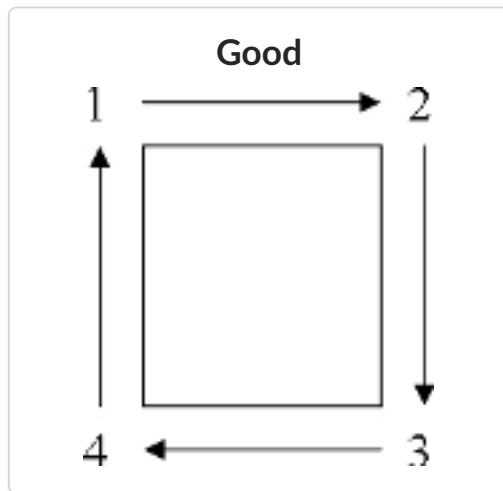
Line type 4 is a filled quadrilateral (also known as a "quad") drawn between four points. The generic format is:

4 <colour> x1 y1 z1 x2 y2 z2 x3 y3 z3 x4 y4 z4

Where:

- <colour> is a number representing the colour of the part. See the [Colours section](#) for allowable colour numbers.
- x1 y1 z1 is the coordinate of the first point
- x2 y2 z2 is the coordinate of the second point
- x3 y3 z3 is the coordinate of the third point
- x4 y4 z4 is the coordinate of the fourth point

While no specific order (winding) is required, the points of the quadrilateral must be declared in either a clockwise (CW) or counter-clockwise (CCW) order (see [BFC statements](#)). The "good" example exhibits clockwise winding. If the points were given in the order 1, 4, 3, 2, they would exhibit counter-clockwise winding and also be considered as "good". The "bad" example illustrates a "bow-tie" or "hour-glass" quad.



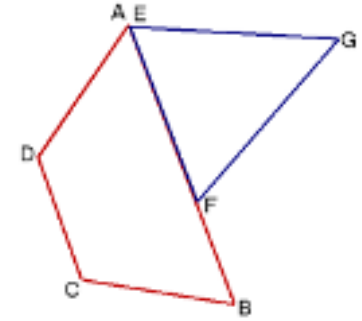
The four points must lie within a plane. Quads should not contain three co-linear points (they should be reduced to a triangle). Quads must not be concave (two or more other polygons must be used instead).

Co-planar polygons (line types 3 and 4) may overlap only if they are the same colour. Any such overlap must be kept to a minimum as it affects the render quality of transparent parts.

Polygons may intersect, but any such intersection must be kept to a minimum as it affects the render quality of transparent parts.

Where possible, abutting polygons should share common vertices. That is, if polygons ABCD and EGF abut along the line AEFB, A and E should be the same point and F and B should be the same point.

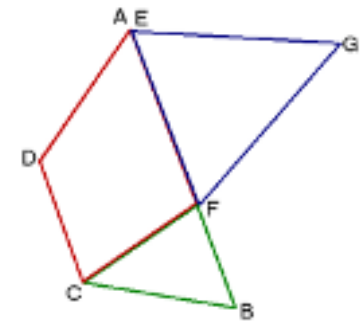
If A and E are the same point but F and B are not the same point, the pixel approximations of lines A'B' and A'F' (where A' represents the pixel point of A - which will be subject to integer truncation) may not be the same along the length A'F'. This will result in holes being visible in the part, especially at high magnification/resolution, and is the primary cause of "dot creep" when rotating parts.



Where practical, if A and E (and/or F and B) are not the same point, one or other or both of the polygons should be decomposed into two or more smaller polygons such that they share common vertices.

In the diagram, ABCD has been decomposed into the quad AFCD and the tri FBC

It is strongly recommended that polygons do not use colour 24 (the complement colour). (The rendition of a colour 24 polygon in a main file is implementation dependant (and may not be rendered at all), while the rendition of a colour 24 polygon in a sub-file is configurable, and so may not give the expected results in all situations/renderers.)



Line Type 5

Line type 5 is an optional line. The generic format is:

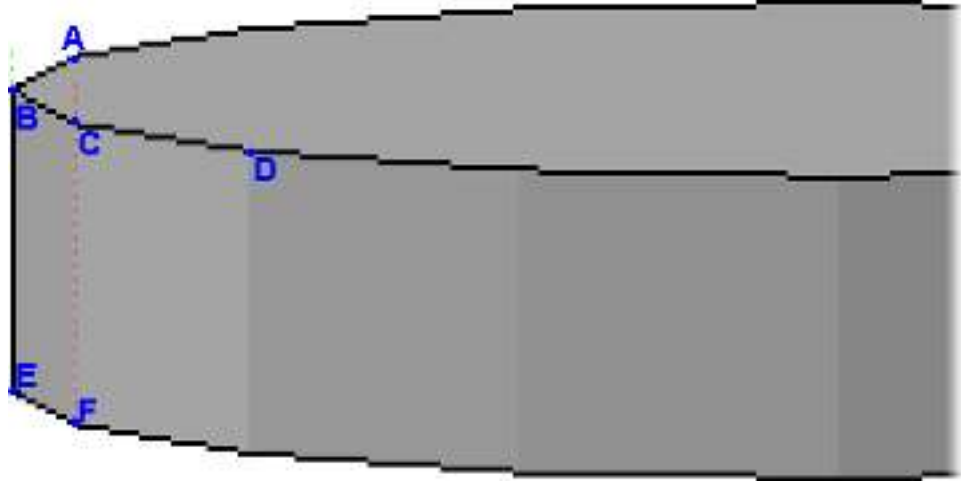
5 <colour> x1 y1 z1 x2 y2 z2 x3 y3 z3 x4 y4 z4

Where:

- <colour> is a number representing the colour of the part, typically this is 24 - the edge colour. See the [Colours section](#) for allowable colour numbers.
- x1 y1 z1 is the coordinate of the first point
- x2 y2 z2 is the coordinate of the second point
- x3 y3 z3 is the coordinate of the first control point
- x4 y4 z4 is the coordinate of the second control point

With an optional line, a line between the first two points will only be drawn if the projections of the last two points (the control points) onto the screen are on the same side of an imaginary line through the projections of the first two points onto the screen.

The optional line can be a difficult concept to grasp. The best way to demonstrate how they work is through a picture.



5 24 Bx By Bz Ex Ey Ez Ax Ay Az Cx Cy Cz

A and C are on the same side of the green line through BE, so BE is drawn.

5 24 Cx Cy Cz Fx Fy Fz Bx By Bz Dx Dy Dz

B and D are *not* on the same side of the red line through CF, so CF is *not* drawn.

This serves to "outline" the edges of the curved surface, which is the intent of optional lines.

As seen above, the control points usually can be chosen from already known points in the object. Since they are never drawn, they can be located anywhere, as long as they have the right controlling properties.

Optional lines have the same colour rules as normal lines (Line Type 2)

Colours

The official colours are defined via Colour Definition statements in the Idconfig.ldr file

See the [LDraw.org Colour Definition Language Extension](https://www.ldraw.org/docs/ldraw.org%20Colour%20Definition%20Language%20Extension) for a description of the Colour Definition Meta statement

The colour numbers 16 and 24 are reserved and must not be used in Colour Definition statements.

Colour 16

Colour 16 is referred to as the "main colour" or "current colour". When a sub-file is referenced by another file all the colour 16 command lines are displayed using the colour of the line that referenced it.

Example: 1 4 0 0 0 1 0 0 0 1 0 0 0 1 part.dat

All the colour 16 command lines in part.dat will be displayed using colour 4

Colour 24

Colour 24 is referred to as the "complement colour" or "edge colour". Colour 24 is most commonly used for line types 2 (line) and 5 (optional line). When a sub-file is referenced by another file all the colour 24 command lines are displayed using the complement colour of the line that referenced it. The complement colour of a given colour is defined in the Colour Definition for that colour.

Example: 1 4 0 0 0 1 0 0 0 1 0 0 0 1 part.dat

All the colour 24 command lines in part.dat will be displayed using the complement colour of colour 4

Use of colour 24 for line types other than 2 and 5 is strongly discouraged.

See the [LDraw Colour Chart](#) for a complete list of the LDraw colour number equivalents to the official LEGO colours.

Direct Colours

Direct Colours are colour numbers that represent RGB values using the form 0x2RRGGBB, where RRGGBB is the exact RGB value in hexadecimal notation. For example, 0x2008000 would produce dark green with RGB of (0 red, 128 green, 0 blue). All letters (A-F) used in the specification of RGB values must be in upper case.

LDraw compatible visualization applications shall be capable of displaying Direct RGB Values from Patterned Parts that contain these colours. For applications that support part authoring, it is recommended that the program provide a graphical aid for choosing a Direct RGB Value. The use of Direct RGB Values is discouraged in model authoring. Therefore, applications that support model authoring should favour colours from LDConfig.ldr.

Blended Colours

LDraw compatible visualization applications that once supported dithered colours should continue to do so in the display of parts and models that may contain a dithered colour code. However, dithered colours have been made obsolete and will not be contained within any new official parts. These colours should be removed from any graphical aids for choosing colours, and should not be added to any new graphical aids for choosing colours. New applications are not required to support these colours.

These "dithered" colours are defined based on a modification of ldraw.exe's original behavior that uses RGB blending instead of dithering. The original ldraw.exe drew them as a 50/50 checkerboard dither of two of its standard 16 colours. Since modern computers all support true colour, they will instead be drawn as a blend of the two input colors, which is calculated as the average of two original ldraw.exe colours.

The two input colour indexes are calculated as follows:

$$n_1 = (n_b - 256) / 16$$

$$n_2 = (n_b - 256) \bmod 16$$

Where:

- n_1 is the index of input colour 1
- n_b is the index of the blended color
- n_2 is the index of input colour 2

Once n_1 and n_2 are determined, the colour of the blended colour is calculated using the following equations:

$$R_b = (R_1 + R_2) / 2$$

$$G_b = (G_1 + G_2) / 2$$

$$B_b = (B_1 + B_2) / 2$$

Where:

- R_b , G_b , B_b are the red, green, and blue components of the blended colour
- R_1 , G_1 , B_1 are the red, green, and blue components of the first input colour
- R_2 , G_2 , B_2 are the red, green, and blue components of the second input colour.

The red, green, and blue components of the original 16 ldraw.exe colours are as follows:

0: 51, 51, 51
 1: 0, 51, 178
 2: 0, 127, 51
 3: 0, 181, 166
 4: 204, 0, 0
 5: 255, 51, 153
 6: 102, 51, 0
 7: 153, 153, 153
 8: 102, 102, 88
 9: 0, 128, 255
 10: 51, 255, 102
 11: 171, 253, 249
 12: 255, 0, 0
 13: 255, 176, 204
 14: 255, 229, 0
 15: 255, 255, 255

Going the other direction, the blended colour's colour code is calculated using the following equation:

$$n_b = n_1 * 16 + n_2 + 256$$

Where:

- n_b is the index of the blended colour
- n_1 is the index of the first input colour
- n_2 is the index of the second input colour

META Commands

Meta commands (or meta statements) are a powerful way to extend the File Format Specification to enable the inclusion of meta data (such as author name, keywords, category, etc) and processing instructions (such as step, bfc, etc) within the data file. Reference the [formal META command definition](#) in the Line Type 0 section

For historical reasons, the original meta commands are defined as part of the file format specification, while others are defined in separate language extension documents.

Header Statements

See the [Official Library Header Specification](#)

STEP

Marks the end of a building step

Format:

0 STEP

The Step command causes the original LDraw program to pause until the Enter key is pressed. Additionally a bitmap of the currently displayed parts in the model is saved to the LDRAW\BITMAP directory. Therefore, for the original LDraw program, 0 STEP is shorthand for 0 SAVE followed by 0 PAUSE

Other programs may interpret this META command differently.

WRITE/PRINT

Displays a message

Format:

0 WRITE <message>

Or

0 PRINT <message>

Where:

<message> is a string of any length.

The Write or Print commands cause the original LDraw program to display a message to the screen. This message is not saved on any bitmaps.

Other programs may interpret this META command differently.

CLEAR

Clears the screen

Format:

0 CLEAR

This command causes all previous drawn sub-files and line types to be cleared (erased). Any following line types are drawn as normal.

PAUSE

Pauses the drawing of a file

Format:

0 PAUSE

This command stops the further parsing of lines until the Enter key is pressed. Unlike the Step command, PAUSE does not save a bitmap of the currently displayed model to the LDRAW\BITMAP directory.

SAVE

Saves a bitmap of the current drawing

Format:

0 SAVE

Saves a bitmap of the current drawing in the LDraw\BITMAP directory. Unlike the Step command, this command does not cause the program to pause.

File Type

Indicates the type of file (model, part, etc), and if it is part of the LDraw.org Parts Library.

Format:

```
0 !LDraw_ORG <type> (qualifier(s)) (update-tag)
```

or

```
0 LDraw_ORG <type> update-tag
```

or

```
0 Official LCAD <type> update-tag
```

or

```
0 Unofficial <type>
```

or

```
0 Un-official <type>
```

The first format (!LDraw_ORG) is the [current standard for files in the official parts library](#). Older files still carry the second and third (Official LCAD) formats. The valid values for the <type> field are:

- All valid values specified in the [Official Library Header Specification](#) or the [Official Model Repository \(OMR\) Specification](#)
- File -
- Submodel -
- Element +
- Sub-part +
- Hi-Res Primitive +
- Alias +
- Cross-reference +

- = only used in unofficial files

+ = obsolete/deprecated, still appears on some files.

Some older official files do not include a type code on their File Type line. Unofficial files may have a variety of values after the initial tag. The exact format of the File Type line has changed over time. In general, parsers should consider this line to be case-insensitive and free-format.

CATEGORY and KEYWORDS Statements

See the [CATEGORY and KEYWORDS Language Extension](#)

MPD Statements

See the [MPD Language Extension](#)

BFC Statements

See the [LDraw Language Extension for Back Face Culling \(BFC\)](#)

Unofficial META Commands

See the [list of known META commands and the programs that use them](#)

Adding New META Commands

The following guidelines should be followed by developers when adding new META command functionality to their LDraw.org compatible programs:

- All new META commands should be based on the program name where this META command is first defined/used. For example, if your program is called LDPARSER and you want to add a META command called EXAMPLE, your command should appear in DAT code as `!LDPARSER EXAMPLE`
- Avoid defining META commands which are similar in name but not equal in function to already existing META command. Doing this will avoid user confusion and possible erroneous output.

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Colour Definition (!COLOUR) Language Extension



Documentation: Colour Definition (!COLOUR) Language Extension

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Maintained by: LDraw.org Standards Board

Revision: 01-Apr-2025

Revision History:

31-Apr-2009 - Original Document

01-Apr-2025 - Added FABRIC material

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Purpose

This meta-statement will specify the properties of a single LDraw colour code. This statement may appear at any time in an LDR file, but will probably be most useful in the file header.

In official LDraw.org materials, !COLOUR will appear in the Idconfig.ldr configuration file, to define the standard LDraw color set. It will not appear in any official part or primitive files.

The current version of Idconfig.ldr is distributed with the [Official Library](#)

The Idconfig.ldr file that contains the standard colour definitions will contain the following in its header:

```
0 !LDRAW_ORG Configuration UPDATE YYYY-MM-DD
```

Scope

It's recommended that colour definitions be limited in scope. A color definition will affect colors from the point it first appears, continuing through the end of the file. Commands preceding a colour definition will not be affected by the definition. Color definitions will expire at the end of the file in which they appear, effectively falling out of scope. Color definitions will be passed to subfiles.

The configuration file, Idconfig.ldr, is not affected by the scoping rules, in the sense that the definitions in Idconfig.ldr remain in effect after the renderer is finished processing Idconfig.ldr.

Syntax

The entire statement must appear on one line. It may be line-wrapped in this document for clarity.

```
0 !COLOUR name CODE x VALUE v EDGE e [ALPHA a] [LUMINANCE l] [ CHROME | PEARLESCENT | RUBBER | MATTE_METALLIC | METAL | MATERIAL <params> ]
```

Tags. CODE, VALUE, EDGE, ALPHA, DITHER, LUMINANCE, CHROME, PEARLESCENT, RUBBER, MATTE_METALLIC, METAL and MATERIAL are keyword tags. Some tags are followed by a single parameter value. Tags are case-insensitive. Numeric parameters, other than RGB color values, should be specified in decimal. RGB values must be specified in hexadecimal, and must be prefixed with # or 0x. Byte values are integers in the range 0-255. Identifiers are strings of text. The allowable characters are alphanumerics from 7-bit ASCII set: a-zA-Z0-9_. Additionally, the first character in an identifier must be alphabetic (a-zA-Z). Identifiers are case-insensitive.

Definitions of Parameters

name

name - A short name for the color. Name may be any identifier. This item is provided mainly for human recognition. Name is a required element.

CODE x

CODE x - The LDraw color code. For LDraw compatibility, x is an integer 0-511. If strict LDraw compatibility is not required, x is any identifier recognized by a rendering program as a color code. Code is a required element.

VALUE v

VALUE v - The color value. v is a 24-bit RGB value. VALUE is a required element.

EDGE e

EDGE e - The contrasting edge color. e may be either a color code or an RGB value. EDGE is a required element.

ALPHA a

ALPHA a - The alpha channel (transparency) value. a is a byte value, where 0 is totally colorless and 255 is completely opaque. In official materials, standard transparent colors will specify ALPHA 128. A 0 value is useful for temporarily hiding sections of models. ALPHA is an optional element.

LUMINANCE l

LUMINANCE l - Brightness for colors that glow. l is a byte value. Luminance is not generally used by LDraw renderers, but may be used for translation to other rendering systems. LUMINANCE is optional.

Finishes

Only one of CHROME, PEARLESCENT, RUBBER, MATTE_METALLIC, METAL and MATERIAL may be specified for a single color. They all specify the finish/texture to be applied to an object being rendered. Other finish-tags may be defined over time.

CHROME

CHROME - A flag indicating that surfaces should have a mirror-like finish. Not currently used by LDraw-compatible programs, but may be implemented in the future. CHROME is optional, and may not appear with other finish tags.

PEARLESCENT

PEARLESCENT - A flag indicating that surfaces should have a finish like Lego's 'metallic' colors. These colors are commonly called 'pearlescent' by fans. Not currently implemented in LDraw-compatible programs. PEARLESCENT is optional, and may not appear with other finish tags.

RUBBER

RUBBER - A flag indicating that surfaces should have a finish like rubber (typically rubber tubing or rubber tyres). RUBBER is optional, and may not appear with other finish tags.

MATTE_METALLIC

MATTE_METALLIC - A flag indicating that surfaces should have a finish that simulates brushed metal finishes. MATTE_METALLIC is optional, and may not appear with other finish tags.

METAL

METAL - A flag indicating that surfaces should have a finish like actual metal (not a simulation of a plastic simulation of metal). METAL is optional, and may not appear with other finish tags.

MATERIAL

MATERIAL [name ...] - Some other surface finish requiring customized parameter(s). Name is an identifier. MATERIAL is optional, and may not appear with other finish tags. Since it may have complex parameters that don't follow the standard syntax of the !COLOUR statement, MATERIAL should be the last tag in the !COLOUR statement.

The following MATERIAL definitions are official:

```
GLITTER VALUE v [ALPHA a] [LUMINANCE l] FRACTION f  
VFRACTION vf SIZE s | MINSIZE min MAXSIZE max  
SPECKLE VALUE v [ALPHA a] [LUMINANCE l] FRACTION f SIZE s  
| MINSIZE min MAXSIZE max  
FABRIC [VELVET | CANVAS | STRING | FUR]
```

GLITTER is used to specify a color that includes glitter mixed into the transparent plastic. Individual glitter grains are assumed to be flat squares, with arbitrary orientation.

SPECKLE is used to specify a color that has a speckled pattern of an alternate color on its surface. Individual speckle grains are assumed to be roughly round.

FABRIC is used to specify a surface finish for fabric. Omitting the second qualifier assumes a finish similar to the fabric used in Minifig capes.

FRACTION f is the portion of the surface that uses the alternate color, expressed as a decimal number greater than 0.0 and less than 1.0. VFRACTION vf is the portion of the volume that uses the alternate color, expressed as a decimal number greater than 0.0 and less than 1.0. Renderers that support volume rendering will use this value instead of the FRACTION value.

SIZE is the size of each speckle/glitter grain, in LDraw units. It must be greater than 0. This is mutually exclusive with MINSIZE and MAXSIZE.

MINSIZE and MAXSIZE provide a range of sizes for grains, in LDraw units. If either of these is specified, both must be specified, and they are mutually exclusive with SIZE.

Here is an example usage of SPECKLE:

```
0 !COLOUR Speckle_Black_Silver CODE 132 VALUE #000000 EDGE
#595959 MATERIAL SPECKLE VALUE #595959 FRACTION 0.4
MINSIZE 1 MAXSIZE 3
```

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Documentation: Back Face Culling (BFC) Language Extension

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Maintained by: [LDraw.org Standards Board](#)

Revision: 09-Nov-2006

Revision History:

09-Nov-2006: Original

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Purpose

To establish a standard for back face culling processing in LDraw-compatible rendering programs. This standard will include language extensions, definitions and expected processing effects.

In this document this standard will be called the BFC extension.

The standard must allow cross-compatible LDraw files. That is, LDraw-compatible rendering programs that don't support the BFC extension must render BFC-enabled LDraw files correctly, while renderers that support the BFC extensions must render non-BFC'ed LDraw files correctly.

To make this standard useful and effective, the LDraw parts library must be updated to follow the new standard. Since it would be difficult to rewrite the entire library in one update, the standard will allow for a mix of BFC enabled and non-enabled files in one rendering.

Definitions

Certified

An LDraw file is certified if it complies with the specifications in this document, includes a 0 BFC CERTIFY meta-statement before the first operational command-line, and does not contain a 0 BFC NOCERTIFY meta-statement.

Clipping

In this document, the term clipping is used as a synonym for culling. While this is technically incorrect, the usage is kept for backwards compatibility with the prior draft and de-facto standards. (In other graphic contexts, clipping usually refers to trimming a geometric primitive to not extend past the edge of a given region, while culling refers to not drawing the primitive at all.)

Culling

In this document, the term culling refers to the process of performing back face culling; that is, not drawing a geometric primitive at all.

File-reference Branch

A set of files in the file-reference tree that either start from the main model file or from a part file. The file-reference tree has a single main branch (the main model) which contains many sub-branches (the parts) which themselves may contain sub-branches (for example, a complete assembly such as a minifig).

Invert

Turning a subfile inside out, usually performed on geometric primitives. When back face culling is used inversion is needed or interior surfaces will never be rendered. Because polygons are one-sided, interior primitives need to have their surfaces face inward, rather than outward.

LDraw file

A single file specifying a part, sub-part, primitive or model. Parts, sub-parts and primitives have the extension DAT, while models can use DAT, LDR or MPD. While BFC meta-statements are rarely used in model files, they still apply, and should be used if the model file contains any polygons.

Operational Command-Line

Any statement in an LDraw file with line-type 1 through 5. In other words, the Subfile, Line, Triangle, Quad, and Conditional Line statements.

Part

An LDraw file which represents a complete real-world building element. These files are stored in the ldraw\parts\ directory.

Polygon

A 2D surface created by LDraw's line-type 3 (triangle) or 4 (quadrilateral) command.

Primitive

An LDraw file, typically small, which models a geometric shape or a standard attribute of building elements, such as studs or the cross-axle shape. These files are kept in the ldraw\p\ directory (for lo-res primitives) and the ldraw\p\48 directory (for hi-res primitives). (In other graphics contexts, the term primitive refers to the basic geometric shapes provided by the rendering environment.)

Subfile

An LDraw file referenced from another file, via a line-type 1 command. Or any file which is lower in the file-reference branch than the current file.

Subpart

An LDraw file which is only a portion of a complete element. It does not necessarily correspond to a discrete portion of a real-world building element. These files are kept in the ldraw\parts\s\ directory.

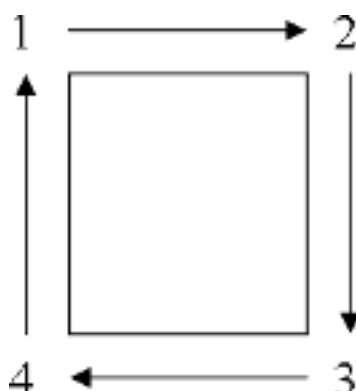
Superfile

The file which referenced the current file. More generally, any file which is higher in the current file-reference branch than the current file.

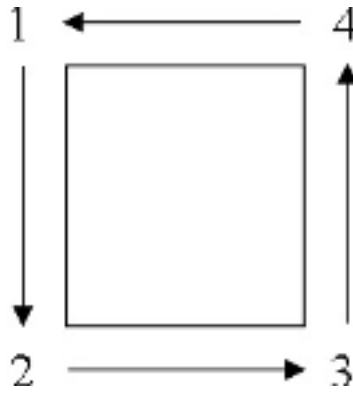
Winding

This is the order vertices are specified in a polygon command. The order can be either clockwise or counter-clockwise (AKA anti-clockwise), and is based on viewing the polygon from its front side.

Clockwise winding (vertices numbered 1 through 4):



Counter-clockwise winding:



Language Extension Functionality

The BFC language extension allows LDraw files to specify and control the following conditions:

Compliance. LDraw files which follow the BFC extension must be clearly and unambiguously marked. It is also useful to allow non-compliant files to be marked as well. Having the compliancy stated plainly simplifies the task of the rendering program, and makes it easier for humans to read files.

Marking a file as BFC-compliant only directly affects that file. In order for subfiles to be treated as compliant, they must also be marked as compliant. Additionally, with the exception of parts, a file is only treated as being BFC-compliant if it and all of its superfiles are compliant. The reason for this is that, while processing, there is no way to know the intended inversion state of a subfile when the file's superfile isn't BFC-compliant. The reason that part files can be excepted from this rule is that they are complex closed solids, so there is never a valid reason to invert them. Assuming part files are never inverted allows the rendering engine to apply BFC-processing on certified parts, even if the calling file(s) (for example, the main model file or sub-models of the main model) aren't certified.

Winding. It must be possible for a file to specify the winding order of the polygon commands within that file: clockwise, counterclockwise, or unknown. Allowing the winding to be set at the file level (as part of the 0 BFC CERTIFY meta-statement) is primarily a convenience for file authors.

Changing the winding setting will only affect the current file. It will not modify the winding of subfiles.

It is permitted to change the winding of polygons within a file. In this case the change of winding direction affects all polygons following the CW/CCW option, either until the end of the current file or another CW/CCW option is encountered.

Culling. It must be possible to enable and disable culling during the processing of a file. But even when culling is currently enabled, it may not be possible to actually cull the polygons. Polygons can be tested for culling only when the following conditions apply:

- All superfiles (in the current file-reference branch) are certified
- The current file is certified
- No superfile has disabled culling prior to referencing this subfile

Unless all of these conditions are met at the time a subfile is rendered, no culling is possible.

If the culling state is modified, it affects all lines following the CLIP/NOCLIP option, either until the end of the current file or another CLIP/NOCLIP option is encountered. When subfiles are referenced, they will receive a flag indicating the accumulated culling state, but there is no sense of a global culling mode.

Inversion. Sometimes, it is desirable to reverse the surfaces of a subfile; to turn the subfile 'inside out'.

One common example of inversion is the cylinder primitive. Cylinder primitives are designed so the surfaces face outward from the center. In 3D tubes, a pair of cylinder primitives are used to model the tube; one outer cylinder is oriented normally, and the inside cylinder, scaled slightly smaller, is required to face inward. This is accomplished by flagging the inner cylinder as being inverted.

Inversion accumulates down the file-reference branch. If the current file is being rendered inverted, then any subfiles of the current file are also rendered as inverted.

Inversion is a boolean operation; inverting a file that is already inverted will give the file the normal orientation. So if the current file is inverted, and a subfile is flagged as inverted, the subfile will be rendered with normal orientation, that is, right-side-out.

(As a practical matter, rendering engines can accomplish inversion simply by switching the order of winding; treat CCW files as CW and vice versa. This must happen in conjunction with the other settings.)

Language Extensions

There is a single meta-statement in the BFC language extension, the 0 BFC statement. The statement includes options to specify BFC-related operations.

Syntax:

```
0 BFC (NOCERTIFY | CERTIFY [CW|{CCW}])
0 BFC ((CW|CCW) | CLIP [(CW|CCW)] | NOCLIP | INVERTNEXT)
```

where (a | b) means either a or b, [] indicates optional items and {} indicates the default value.

This syntax results in the following permissible variations of the BFC meta-statement

```
0 BFC NOCERTIFY
```

```
0 BFC CERTIFY (CCW is implied)
```

```
0 BFC CERTIFY CW
```

```
0 BFC CERTIFY CCW
```

```
0 BFC CW
```

```
0 BFC CCW
```

```
0 BFC CLIP (winding is unchanged)
```

```
0 BFC CLIP CW
```

```
0 BFC CLIP CCW
```

0 BFC NOCLIP

0 BFC INVERTNEXT

The BFC meta-statement and all of its options are case-sensitive, ie, they must always be in all capitals.

The BFC meta-statement shall ignore repeating whitespace, and accept any number of spaces or tabs as equivalent to a single space.

In order for a file to be processed with back face culling, there must be at least one 0 BFC meta-statement before the first operational command-line. If there is no such 0 BFC meta-statement in the file, BFC processing will be disabled for that file.

Only one NOCERTIFY/CERTIFY meta-statement may be present in a file, and if present, must precede all other BFC meta-statements and operational command-lines.

All BFC commands that act on succeeding lines in the file shall ignore empty lines.

For backwards compatibility the CLIP and winding options may be specified in any order on the meta-statement, so the following are also valid

0 BFC CW CLIP

0 BFC CCW CLIP

Language Operations

CERTIFY

This option indicates the LDraw file is compatible with the back face culling extension. Every LDraw file must be clearly labeled if it is compliant. One way to accomplish this is to place 0 BFC CERTIFY at the beginning of the file, before the first operational command-line.

A second way to specify a file as compliant is to use any option, except for NOCERTIFY, on a 0 BFC meta-statement, before the first operational command-line. This is an acceptable alternative, but the 0 BFC CERTIFY method is recommended and preferred.

Files in the LDraw.org Parts Library, if they are BFC compliant, are required to have an explicit 0 BFC CERTIFY line in their header.

If a file has no 0 BFC meta-statement before the first operational command-line, 0 BFC NOCERTIFY is assumed and BFC processing will be disabled for the file.

NOCERTIFY

This option specifies that the file is not compliant with the BFC specification; the polygons are not wound consistently and/or correctly, and/or subfile references are not properly inverted. If the NOCERTIFY option is used, it must appear before any operational command-lines. Any other BFC meta-statements in the file will be ignored.

CLIP

This option sets the cull-state to enabled. This allows culling, if all other conditions for culling are met. (While this option should be called CULL, the CLIP form is intentionally being kept for backwards compatibility.)

NOCLIP

This option sets the cull-state to disabled. Any subpart or primitive referenced while the cull-state is disabled will not be eligible for culling. (While this option should be called NOCULL, the NOCLIP form is intentionally being kept for backwards compatibility.)

CLIP/NOCLIP Notes

There may be any number of changes to the cull-state in a file, although it is recommended that such changes be kept to a minimum.

If neither the CLIP nor NOCLIP option is specified on a 0 BFC meta-statement before the first operational command-line, then that file's cull-state is set to enabled (CLIP).

CW

This option sets polygon winding to clockwise.

CCW

This option sets polygon winding to counter-clockwise.

CW/CCW Notes

There may be any number of changes to the winding direction in a file, although it is recommended that changes to winding are kept to a minimum.

If no winding option is specified for a file, the local winding state will be defaulted to counter-clockwise (CCW).

INVERTNEXT

This option inverts a subpart or primitive. It may only be used immediately before a subfile command line [while intervening blank lines are permitted, they are discouraged for reasons of readability], and it only influences the immediately following subfile command. It should never be used before a part.

Example:

```
0 BFC INVERTNEXT
1 16 0 0 0 1 0 0 0 1 0 0 0 1 somefile.dat
1 16 0 0 0 1 0 0 0 1 0 0 0 1 another.dat
```

In this example, somefile.dat would be rendered as inverted while another.dat would not be inverted.

For further information, see "Inversion" in the Language Extension Functionality section.

LDraw Parts Library Guidelines

(This section gives some suggestions for implementing BFC within the LDraw Parts Library and the impact on the current library. This section may be varied by other specifications (for example, the Parts Library Header specification)

and/or the requirements of the Parts Library administrators without affecting the validity of the Back Face Culling specification.)

The LDraw Parts Library includes all parts, primitives, and subparts distributed with LDraw or in an ldraw.org parts update.

New parts will not be required to be compliant with this extension. They will however be required to carry a 0 BFC meta-statement, indicating either compliance or non-compliance.

New primitives will be required to be certified before being accepted for release.

It is desirable for all files to use the same winding. When possible, files should use counter-clockwise winding. The actual winding for any part is left to the file author. Primitives should always use CCW winding.

Primitives should generally be written so that polygons face outward, or upward. Exceptions are allowed for polygons which model inward- or downward-facing surfaces.

As noted in the language sections, all BFC compliant files in the parts library will have an explicit 0 BFC CERTIFY line in their header.

Rendering Engine Guidelines

(This section gives some suggestions for the design of programs in order to achieve correct renderings. Any program may violate these guidelines if there is another way to achieve a valid rendering.)

Matrix Reversals. Rendering engines will need to correct for orientation matrices which inadvertently or deliberately reverse a subfile.

Normal transformation:

```
1 16 0 0 0 1 0 0 0 1 0 0 0 1 somefile.dat
```

'Reversed' transformation:

```
1 16 0 0 0 1 0 0 0 -1 0 0 0 1 somefile.dat
```

If the rendering engine does not detect and adjust for reversed matrices, the winding of all polygons in the subfile will be switched, causing the subfile to be rendered incorrectly.

The typical method of determining that an orientation matrix is reversed is to calculate the determinant of the matrix. If the determinant is negative, then the matrix has been reversed.

The typical way to adjust for matrix reversals is to switch the expected winding of the polygon vertices. That is, if the file specifies the winding as CW and the orientation matrix is reversed, the rendering program would proceed as if the winding is CCW.

The INVERTNEXT option also reverses the winding of the polygons within the subpart or primitive. If the matrix applied to the subpart or primitive has itself been reversed the INVERTNEXT processing is done IN ADDITION TO the automatic inversion - the two effectively cancelling each other out.

Inverted Subfiles. Generally, it is not possible to determine that a subfile reference is inverted or normal (which is the reason for the 0 BFC INVERTNEXT meta-statement). In particular, the rendering engine should *not* use the determinant of the orientation matrix to determine if the subfile is intended to be inverted (see 'Matrix Reversals', above).

Note: Part files are never inverted as they represent closed solids.

Uncertified Files. No assumptions can be made about models which make direct use of primitives or polygon commands, so a rendering engine should not simply treat uncertified model files as certified.

Culling State. The rendering engine can default to either allow or disable culling at the start of processing. Presumably, the user will be given the ability to control this state.

Degenerate Matrices. Some orientation matrices do not allow calculation of a determinate. This calculation is central to BFC processing. If an orientation matrix for a subfile is degenerate, then culling will not be possible for that subfile.

Rendering Processing

(This section presents a possible model for writing the core processing loop in an LDraw/BFC rendering program. Any program may violate this model if there is another way to achieve a valid rendering.)

All BFC-relevant logic is included while as much other logic (as possible) is excluded. It should not be assumed that this pseudo-code represents the most efficient way to implement BFC.

The function BFC() returns a boolean value, indicating whether a polygon should be rendered or culled. As the nature of this routine does not impact the BFC standard, the logic for BFC() is not included in the following pseudo-code. There is information about BFC processing available from many locations, including lugnet.cad.dev

Recursive Procedure RenderFile

Parameters:

```
ModelFile string           // File to render
AccumCull boolean          // global culling value yes/no
AccumInvert boolean        // current inversion
normal/inverted
AccumTransformMatrix matrix // current transformation
Colour integer             // current colour
```

Declare

```
LocalCull  boolean          Initial TRUE
Winding    bvalue(CCW, CW)  Initial CCW
Certified  trivalued(TRUE, FALSE, UNKNOWN) Initial UNKNOWN
InvertNext boolean          Initial FALSE
Command    DATCommandLine // Structure containing parameters
```

from

```
// a single operational command-line.
```

OpenFile(ModelFile)

Do Until EOF(ModelFile)

```

Get Next Command
If Command.Colour = 16 Then
    Command.Colour = Colour
ElseIf Command.Colour = 24 Then
    Command.Colour = EdgeColour(Colour)
End If

Case Command.LineType
    BFC
        If Certified is UNKNOWN and no Option in Command is
NOCERTIFY Then
            Certified = TRUE
        End If

        Do for each Option in Command
            Case Option
                CERTIFY
                    Assert Certified != FALSE
                    // Triggers error if file has been
NOCERTIFY'ed
                    Certified = TRUE
                NOCERTIFY
                    Assert Certified != TRUE
                    // Triggers error if file has been CERTIFY'ed
                    Certified = FALSE
                CLIP: LocalCull = TRUE
                NOCLIP: LocalCull = FALSE
                CCW
                    If AccumInvert Then
                        Winding = CW
                    Else
                        Winding = CCW
                CW
                    If AccumInvert Then
                        Winding = CCW
                    Else
                        Winding = CW
                INVERTNEXT
                    InvertNext = TRUE
            End Case
        End Do
    SUBFILE
        If Certified is UNKNOWN Then
            Certified = FALSE
        Case Certified
            TRUE
                RenderFile Command.Subfile,
                    (AccumCull and LocalCull),
                    (AccumInvert xor InvertNext),
                    Command.TransformMatrix *
AccumTransformMatrix,
                    Command.Colour
            FALSE, UNKNOWN
                RenderFile Command.Subfile,
                    FALSE,
                    (AccumInvert xor InvertNext),
                    Command.TransformMatrix *

```

```

AccumTransformMatrix,
    Command.Colour
LINE, CONDITIONAL_LINE
    If Certified is UNKNOWN Then
        Certified = FALSE
    Deal with primitive command
TRIANGLE, QUAD
    If Certified is UNKNOWN Then
        Certified = FALSE
    End If
    If AccumCull and LocalCull And (Certified is TRUE) Then
        If BFC(Command, AccumTransformMatrix, Winding) Then
            Render Command
        Else
            Don't render Command
        End If
    Else
        Render Command
    End If
End Case

If Command.LineType != BFC Then
    InvertNext = FALSE
ElseIf No Option in Command is INVERTNEXT Then
    InvertNext = FALSE
End If
Loop
End Procedure

```

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Texture Mapping (!TEXMAP) Language Extension

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Texture Mapping (!TEXMAP) Language Extension



Documentation: Texture Mapping (!TEXMAP) Language Extension

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Maintained by: LDraw.org Standards Board

Revision: 01-Aug-2020

Revision History:

30-Oct-2012 - Original

01-Aug-2020 - Clarified fallback section and added search path examples

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Purpose

This document describes the !TEXMAP meta-statement, which contains a set of commands for including texture maps in LDraw files.

Syntax

```
0 !TEXMAP (START | NEXT) <method> <parameters> <pngfile>
[GLOSSMAP pngfile]
```

```
0 !: <geometry1>
```

```
...
```

```
<geometry2>
```

```
...
```

```
0 !TEXMAP FALLBACK
```

```
<geometry3>
```

```
...
```

```
0 !TEXMAP END
```

```
!:
```

```
!:
```

This is used to specify texture mapped geometry that will be ignored by renderers that do not support the !TEXMAP meta-statement. This command must not be nested inside itself. Note: the actual meta-command here is the colon. This was carefully chosen to be brief, while not conflicting with existing

software (since : isn't an existing meta-command). It also makes reading the files easier.

!TEXMAP

This is the main texture mapping meta-command.

The START command indicates that the given texture should be used. (If another texture is currently in use, it is pushed onto a stack for retrieval when an END command is given.) This texture will remain in effect until:

- An END is given within the current file
- The file in which the START is located ends
- A STEP command is reached

The texture will remain active when processing an included file unless overridden within that file.

The FALLBACK command is a section marker. In a TEXMAP supporting application, all geometry lines (whether following the 0 !: comment marker, or presented as standard LDRAW geometry: geometry1 or geometry2 lines) between START and FALLBACK are to be used as geometry and have the texture applied. All geometry between FALLBACK and END are ignored. In applications that don't support TEXMAP, nothing needs to be done, and all non-commented geometry is generated as normal (though the LDRAW parts author should be very cautious not to introduce unintended non-commented geometry lines into the START to FALLBACK section).

Note: the FALLBACK section is always optional, although it is highly recommended any time there is no geometry2 section.

In general, the START to FALLBACK section will contain only geometry following the special comment marker 0 !:, and geometry between FALLBACK and END will not.

NOTE: be cautious (and think through your approach) when including geometry2 lines. This feature is present for use generally in the case where the textured and non-textured FALLBACK geometry are the same. This is a rare occurrence, and normally would not be used.

The END command stops using the current texture and restores the previous texture to current status. This means that nested commands with START will form a stack of textures and the END command will pop those textures. However, END cannot appear in a different file from START. If an END command is given in a sub file without a START having been given in that same file, the END has no effect.

The NEXT command is a shortcut to make a texture work only for the next 1, 2, 3, 4, or 5 line. It would be equivalent to use the START command and then to place an END command immediately after the next 1, 2, 3, 4, or 5 line (with no FALLBACK possible). The first non-blank line following this command **must not** be line type 0. If it is, the NEXT command itself should be ignored (and an error should be generated, if in an environment that reports errors to the user).

The method token specifies the projection method to use for the texture map. Each projection method has its own unique set of parameters, and parameter parsing must be based on the method. Projection methods (and their parameters) are described below.

The first PNG file represents the texture bitmap to apply. The PNG can be B&W or color and if an alpha channel exists, it should be applied appropriately (allowing the part color to show through). The second optional PNG is used as a glossmap. It should be a single channel image where the value indicates the amount of specularity to add at the part of the texture map (RG and B are currently ignored – but reserved – in gloss maps, and the A (alpha) channel determines the amount of gloss at a given texel).

File names that contain spaces must be enclosed in double quotes ("). Additionally, a search for the specified texture file will first be attempted after adding a "textures/" prefix to the filename and following the standard LDraw search path. If no file is found, the search will be repeated without the "textures/" prefix. Filenames that themselves contain double quotes must use \ as an escape character immediately prior to the double quote. The \ character itself must be doubled up (\\) in order to be used to specify a sub-directory, but it is recommended that / be used instead in this case. (In LDraw files, \ and / are interchangeable as path separators.)

Search path examples:

0 !TEXMAP ... 123.png will look for textures/123.png then 123.png
0 !TEXMAP ... s/123.png will look for textures/s/123.png then s/123.png

Projection Methods

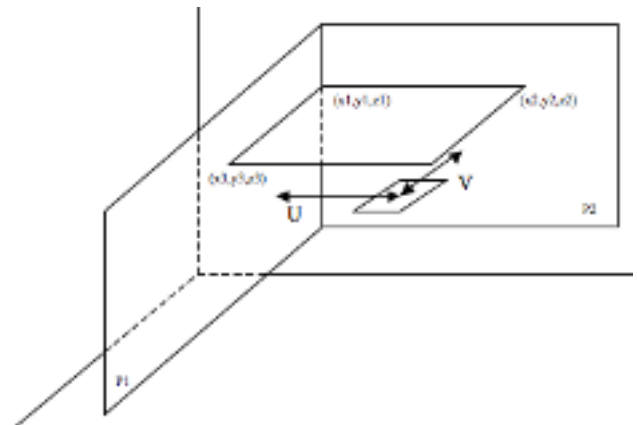
There are three possible methods that can be used to apply textures: PLANAR, CYLINDRICAL, and SPHERICAL.

PLANAR

The parameters for PLANAR are:

x1 y1 z1 x2 y2 z2
x3 y3 z3

These three points represent 3 corners of the texture map in world coordinates. They form a plane onto which objects can be projected and they also represent the extents of the texture.



In addition, points 1 and 2 represent a plane (P1) perpendicular to the texture where point 1 lies on the plane and the normal to the plane follows the line from point 1 to 2. Similarly points 1 and 3 represent a plane (P2) which is perpendicular to both planes. This plane also contains point 1 but its normal follows the line from point 1 to 3.

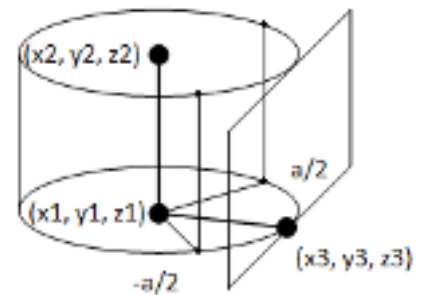
Now to map from world coordinates to texture coordinates, U is given by the distance between a point and plane P1 divided by the length of the vector from point 1 to point 2. V is given by the distance between a point and plane P2 divided by the length of the vector from point 1 to point 3. Now U and V are values normalized to between 0 and 1. These can now be mapped to the pixels of the texture map.

CYLINDRICAL

The parameters for CYLINDRICAL are:

$x_1 \ y_1 \ z_1 \ x_2 \ y_2 \ z_2 \ x_3 \ y_3 \ z_3 \ a$

The first point represents the bottom center of the cylinder. The second point represents the top center of the cylinder. The third point represents a location on the outer edge of the cylinder bottom where the center-bottom of the texture would touch.



The angle indicates the how much of the cylinder is mapped by the texture and so the extents would be $-a/2$ to $a/2$ as measured relative to the radial line from point 1 to point 3.

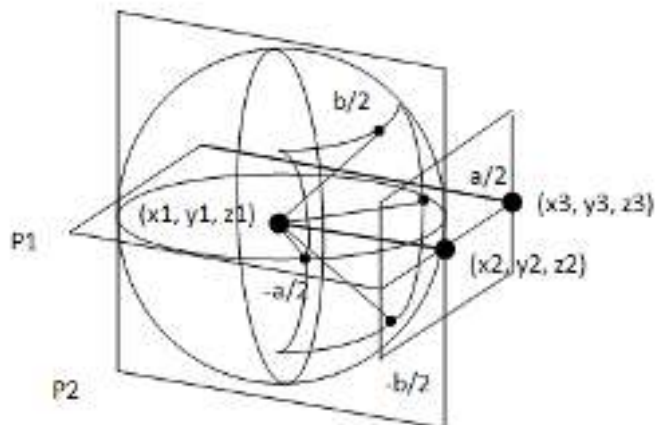
Now to map from world coordinates to texture coordinates, U is given by the angle between the vector formed by points 1 and 3 and the vector formed by point 1 and a world point that has been projected to the plane in which the base of the cylinder occupies. This angle is then divided by the "a" parameter to normalize it to the 0 to 1 range. V is given as the distance of a point from the plane occupied by the base of the cylinder and divided by the length of the vector from point 1 to point 2.

SPHERICAL

The parameters for SPHERICAL are:

$x_1 \ y_1 \ z_1 \ x_2 \ y_2 \ z_2 \ x_3 \ y_3 \ z_3 \ a \ b$

The first point represents the center of the sphere. The second point represents a point on the sphere where the center of the texture map will touch. The



third point is used to form a plane (P1) that is perpendicular to the texture and bisects it horizontally. An additional plane (P2) can be computed by using points 1 and 2 and generating a 3rd point along the normal of P1. P2 will be perpendicular to both P1 and the texture and will bisect the texture vertically. The two angles indicate the extents of the sphere that get mapped to. These are $-a/2$ to $a/2$ and $-b/2$ to $b/2$ as measured relative to the vector from point 1 to point 2 and within the planes P1 and P2 respectively.

Now to map world coordinates to texture coordinates, U is given by the angle between the vector formed by points 1 and 2 and the vector formed by point 1 and a world point that has been projected to the plane P1. The angle is divided by "a" to normalize it to between 0 and 1. V is given by the angle between the vector formed by points 1 and 2 and the vector formed by point 1 and a world point that has been projected to the plane P2. The angle is divided by "b" to normalize it.

- Purpose
- Syntax
 - !:
 - !TEXMAP
- Projection Methods
 - PLANAR
 - CYLINDRICAL
 - SPHERICAL

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!CATEGORY and !KEYWORDS Language Extension

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!CATEGORY and !KEYWORDS Language Extension



Documentation: !CATEGORY and !KEYWORDS Language Extension

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Maintained by: LDraw.org Standards Committee

Revision: 1.7, 18-Nov-2022

Revision History:

1.5 Changed Propellor to Propeller 1.6 Added Duplo 1.7 Added Moved, Helper, and String. Clarified use for Helper, Moved, and Obsolete.

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Purpose

Outlines the !CATEGORY and !KEYWORDS Meta statements that provide information for organizing and searching the part library.

Scope

The !CATEGORY and !KEYWORDS meta-statements provide information for organizing and searching the LDraw parts library. Before these meta-statements were defined, the only way to organize the library was by the descriptive name and part number. While useful, the name and number do not provide enough information to thoroughly catalog a large library of parts.

The !CATEGORY meta-statement provides a means to place the part in a comprehensive classification system and is used for locating a part within a list of categories, while the !KEYWORDS meta-statement provides a means for associating free-form search terms with the part.

The !CATEGORY meta-statement is used exclusively in part files and subparts; it does not appear in models or primitives.

The !KEYWORDS meta-statement is used exclusively in part files and models; it does not appear in subparts or primitives.

!CATEGORY Syntax

0 !CATEGORY <categoryname>

The !CATEGORY meta-statement, if present, must appear in the part header, as defined by the [LDraw.org Official Library Header Specification](#). A part can

have only one !CATEGORY meta-statement. If a part has no !CATEGORY meta-statement, the category is assumed to be the first word in the part name (ie. the first word of the first line of the part file).

The text after !CATEGORY is treated as a single category name. So, if a part contained the meta-statement "`0 !CATEGORY Minifig Accessory`" it would specify that the part belongs in the category "Minifig Accessory" (as opposed to belonging to the "Accessory" subcategory of the "Minifig" category or both the "Minifig" and "Accessory" categories).

Approved !CATEGORY list

New categories are added by the Parts Library administrators. The current list of approved categories is:

- Animal
- Antenna
- Arch
- Arm
- Bar
- Baseplate
- Belville
- Boat
- Bracket
- Brick
- Car
- Clikits
- Cockpit
- Cone
- Constraction
- Constraction Accessory
- Container
- Conveyor
- Crane
- Cylinder
- Dish
- Door
- Duplo
- Electric
- Exhaust
- Fence
- Figure
- Figure Accessory
- Flag
- Flexible
- Forklift
- Freestyle
- Garage
- Glass
- Grab
- Helper
- Hinge
- Homemaker
- Hose
- Ladder
- Lever
- Magnet

- Minifig
- Minifig Accessory
- Minifig Footwear
- Minifig Headwear
- Minifig Hipwear
- Minifig Neckwear
- Monorail
- Modulex
- Moved
- Mursten
- Obsolete
- Panel
- Plane
- Plant
- Plate
- Platform
- Pov-RAY
- Propeller
- Quatro
- Rack
- Roadsign
- Rock
- Scala
- Screw
- Sheet Cardboard
- Sheet Fabric
- Sheet Plastic
- Slope
- Sphere
- Staircase
- Sticker
- Sticker Shortcut
- String
- Support
- Tail
- Tap
- Technic
- Tile
- Tipper
- Tractor
- Trailer
- Train
- Turntable
- Tyre
- Vehicle
- Wedge
- Wheel
- Winch
- Window
- Windscreen
- Wing
- Znap

!KEYWORDS Syntax

0 !KEYWORDS <first keyword>, <second keyword>, ..., <last keyword>

The !KEYWORDS meta-statement(s), if present, must appear in the part header, as defined by the [LDraw.org Official Library Header Specification](#). A part may have many !KEYWORDS meta-statements.

The list of words and phrases following the !KEYWORDS meta-statement is delimited by commas. Any white-space should be reduced to a single space, and leading and trailing spaces on the keyword ignored. Multiple lines can be used to structure or group keywords into themes. The lines should be limited to 80 characters. Words and terms from the part name are not allowed; neither is the part number.

The keywords are a list of words and phrases that further describe the part and specify any terms the author feels would aid in identifying the part. Keywords typically include: terms not included in the part's descriptive name; geometric terms; synonyms (for example, if the part name includes the term "rounded", a keyword might be "curved"); the unique set the part appeared in; LEGO official playthemes; other category names that the part is similar to (for example, wheels without separate tyres could also have the keyword "Tyre"); alternate spellings.

Examples

0 Landing Skids

0 Name: 30248.dat

...

0 !CATEGORY Plane

0 !KEYWORDS helicopter, skid, rail, Arctic, sled, ski, sleigh

0 !KEYWORDS Belville, Town

0 Tile 1 x 2 with Playing Cards Pattern

0 Name: 3069bpw1.dat

...

0 !KEYWORDS western, wild west, spaghetti western, horse opera, cowboy

0 !KEYWORDS bandit, desperado, cardsharp, swindler

0 !KEYWORDS casino, bridge, blackjack, twenty-one, baccarat, chemin de fer

0 !KEYWORDS poker, stud, stud poker, strip poker, straight poker, penny ante

0 !KEYWORDS penny ante poker, high-low, draw, draw poker, pair, straight

0 !KEYWORDS flush, full house, straight flush, royal flush, eight-card stud

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Multi-Part Document (MPD) and Image Embedding (!DATA) Language Extension

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Multi-Part Document (MPD) and Image Embedding (!DATA) Language Extension



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Maintained by: LDraw.org Standards Board

Revision: Revision: 2, 26-May-2020

Revision History:

Revision 1: Original

Revision 2: Added !DATA meta

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Purpose

MPD files or "Multi-Part Documents" are a way to combine several LDraw and encoded binary files into one consolidated source. This allows for ease in posting or emailing a model made up of many subparts.

Structure

A MPD file consists out of blocks of LDraw code separated by 0 FILE or 0 !DATA statements. Each block is considered a separate file and can be referenced by the other ones as needed.

Blocks starting with the 0 FILE statement are normal type 0..5 line LDraw code.

Blocks starting with the 0 !DATA statement contain binary data encoded using a multitude of 0 !: lines. Each of those lines contain a chunk of base64 encoded data which a parser must combine before decoding. No other LDraw statements may be used inside a !DATA block.

The end of each block, or just the last block in the MPD, may be marked with a 0 NOFILE line. The 0 NOFILE command is only required if the file's contents are followed by non-LDraw content (such as the poster's signature lines). LDraw parsers must then ignore all content until a new 0 FILE or 0 !DATA is found.

In order to support the inclusion of MPD files in message systems (like email), any text lines before the first 0 FILE or 0 !DATA statement will be discarded. It

is considered to be an error for any LDraw code (other than comment lines) to appear before the first 0 FILE or 0 !DATA statement.

The first block in the MPD is treated as the 'main model' -- all other files in the MPD will only be rendered if they are referenced by the main model, directly or indirectly. For this reason it is recommended (but not required) to always use a 0 FILE block as the first entry in the MPD.

So far, there are no clear scoping or namespace rules on MPD files. If you put a file named stud.dat in your MPD file, don't be surprised to see your stud.dat file appear on the top of every single brick in your scene.

MPD META Statements

FILE

Format: 0 FILE <filename>

Where:

<filename> is the name of the following LDraw file.

!DATA

Format: 0 !DATA <filename>

Where:

<filename> is the filename of the following encoded data.

!:

Format: 0 !: <base64 string>

Where:

<base64 string> is a chunk of the base64 encoded data (A-Za-z0-9+/) with a maximum length of 80 characters (60 bytes of data). Each line except the last should use the same length and be a multiple of 4 characters. Padding the last line with '=' characters to complete its last group of 4 is optional.

NOFILE

Format: 0 NOFILE

There are no options or parameters.

Example

```
0 FILE main.ldr
1 7 0 0 0 1 0 0 0 1 0 0 0 1 819.dat
1 4 80 -8 70 1 0 0 0 1 0 0 0 1 house.ldr
1 4 -70 -8 20 0 0 -1 0 1 0 1 0 0 house.ldr
1 4 50 -8 -20 0 0 -1 0 1 0 1 0 0 house.ldr
1 4 0 -8 -30 1 0 0 0 1 0 0 0 1 house.ldr
1 4 -20 -8 70 1 0 0 0 1 0 0 0 1 house.ldr

0 FILE house.ldr
1 16 0 0 0 1 0 0 0 1 0 0 0 1 3023.dat
1 16 0 -24 0 1 0 0 0 1 0 0 0 1 3065.dat
1 16 0 -48 0 1 0 0 0 1 0 0 0 1 3065.dat
1 16 0 -72 0 0 0 -1 0 1 0 1 0 0 3044b.dat
1 4 0 -22 -10 1 0 0 0 0 -1 0 1 0 sticker.ldr

0 FILE sticker.ldr
0 UNOFFICIAL PART
0 BFC CERTIFY CCW
```

```

1 16    0 -0.25 0    20 0 0    0 0.25 0    0 0 30    box5.dat
0 !TEXMAP START PLANAR    -20 -0.25 30    20 -0.25 30    -20 -0.25
-30    sticker.png
4 16    -20 -0.25 30    -20 -0.25 -30    20 -0.25 -30    20 -0.25 30
0 !TEXMAP END

0 !DATA sticker.png
0 !:
iVBORw0KGgoAAAANSUgAAAFAAAAAB4CAIAAADqj0KhAAAAAXNSR0IArs4c6QAAAA
RnQU1BAACx
0 !:
jwv8YQUAAAAJcEhZcwAADsMAAA7DAcdvqGQAAAEUSURBVHhe7du9DcIwFABhk5WgQL
SsQM0UjMEU
0 !:
1BQsQIsoYAt6NkAYxQV/JQ7WvfuKkFTR6Um0FJzR9bJLkXTlNwyD6QymM5ju5Tl8m6
7KGUt3XJcz
0 !:
J/yY8HZ/6C8BFvNZPoesMF0BtMZTGcwncF0BtMZTGcwncF0BtMZTGcwnf8t0bmLh8
5g0oPpDKYz
0 !:
mM5g0oPpDKYzmM5gunDBf3tN+/zqNKt367cb0eGUTstxf1nJZHP0x68T/u3XB5/7/z
MXLTqD6Qym
0 !:
M5j0YDqD6QymM5j0YDqD6QymM5j0YDqD6QymM5j0YLpwwW3t8ajBXTxtTHgwLlp0Bt
MZTGcwncF0
0 !: BtMZTNfKZzyDiT3hCFy06IIFp3QH/CBMh66aBy4AAAAASUVORK5CYII=

```

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Documentation: Localisation Guideline

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Maintained by: LDraw.org Standards Board

Revision: 06-Jan-2010

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Specification

A guideline for implementing localisations for the LDraw System of Tools.

Format

Encoding

The translation file has to be stored encoded by UTF-8.

Line Termination

Line Termination of this file is CrLf. Please see: [LDraw.org File Format Restrictions for the Official Library](#).

Directory

The translations files are stored in the following folder:

```
<LDRAWBASEDIR>\localisations\<language ISO 639-1 in lower case>[-<country ISO 3166-1 alpha-2 in upper case>]
```

Please note the dash between the language and the optional country code. The optional country code is only needed if there is more than one translations for a language that depends on the region where that language is used (For example German or English language, that is used also in other countries where it is slightly different but still that language.) In those cases the most common idiom is used as the default language without country code and the additionally language is marked additionally with the country code. (Example: Language is German, if there will be then the Swiss German translation added the German-Germany translation is marked with de and the German-Swiss translation is marked with de-CH)

So if then someone wants to have the German-Austria translation (de-AT) and that is not available the user gets the de translation instead. If there is also no translation in the de the native language of LDraw.org is used (Australian

English).

Because of the native language of LDraw.org (Australian English) it is necessary for the English language to carry in every case the country code and there should not be any en folder in the localisations folder, only en-<country code> labeled folder.

The translations for the part descriptions are stored in a file called "parts.txt"

The translations for the colour names are stored in a file called "colours.txt"

The translations for the category names are stored in a file called "categories.txt"

The keywords for each part are stored in a file called "keywords.txt"

Header

Header of the translation files:

```
0 LDraw.org Configuration File
0 Name: colours.txt
0 Author: LDraw.org
0 !LDRAW_ORG Configuration UPDATE 2009-09-05
0 !TRANSLATION COLOURS de-DE
0 !LICENSE Redistributable under CCAL version 2.0 : see
CAreadme.txt
```

Please note that also other comment lines can be added such as 0 !HISTORY to reflect the changes that has been made.

All lines that are not starting with a 0 (zero) will be treated as an translation entry that has to follow the rules described below: The first four lines are described in [LDraw.org Official Library Header Specification](#) and are adopted for these files as follows:

```
0 LDraw.org Configuration File - Always this text if official
document.
0 Name: colours.txt - filename not limited to dat or ldr
0 Author: LDraw.org - LDraw.org if official document. Can be also
another author name.
0 !LDRAW_ORG Configuration UPDATE 2009-09-05 - See: LDraw.org
Colour Definition Language Extension
```

If it is an official LDraw.org document then a line with the following content is necessary:

```
0 !LICENSE Redistributable under CCAL version 2.0 : see
CAreadme.txt
```

Translation Files

The following line applies for translation files and will be described below:

```
0 !TRANSLATION <type of content> <language ISO 639-1 in
lower case>[-<country ISO 3166-1 alpha-2 in upper case>]
```

(Please note the dash between the language and the country code. Country code is optional, please see above.)

Where:

- <type of content> can have the following content COLOURS, PARTS, CATEGORIES, KEYWORDS but is not limited to that. This is also the name of the file in upper case whereas the filename itself is always lower case!
- <language ISO 639-1 in lower case> - See http://en.wikipedia.org/wiki/List_of_ISO_639-1_codes for valid codes.
- <country ISO 3166-1 alpha-2 in upper case> - See http://en.wikipedia.org/wiki/ISO_3166-1_alpha-2 for valid codes.

Type of content

Translation section for COLOURS, PARTS, CATEGORIES and KEYWORDS

"<neutral language>" = "<translation language>"

Where:

- <neutral language> is in any case Australian english. This is also the fallback language if a translation is not found (language used in the part files itself). If a human readable version of the neutral language is needed then add a version of en-AU to your localisation folder. Please note that the keywords are translated keyword by keyword ([LDraw.org CATEGORY and KEYWORDS Language Extension](#))
- <translation language> is the language determined by the <language short letter according ISO 639-1 in lower case>[-<country ISO 3166 in upper case>].

Both strings have to be surrounded by " (doublequote). One or more whitespaces (space, tab) outside of the so delimited strings are allowed. There is no length restriction for these lines. If the string itself contains one of the following characters " (doublequote) or \ (backslash) that characters needs a leading \ (backslash). A single \ (backslash) found in the file will be not processed. It will be deleted if not in front of " (doublequote) or \ (backslash).

Example for this: Content of the file: "Brick with \"=\\" Pattern" will be parsed to "Brick with "=" Pattern"

Example lines: "Brick 1 x 1 with Blue \"0\\" Pattern" = "Stein 1 x 1 mit Blau \"0\\" Dekor"

As the translations strings are only 'virtual' and not coded in the file there are no other restrictions.

Special

To make life easier for translators the following key can be used for ~Moved to files but it is not required.

"~Moved to %s" = "[...] %s [...]"

Application should ideally support both ways.

Examples

Here a sample for the colours.txt:

```
0 LDraw.org Configuration File
0 Name: colours.txt
0 Author: LDraw.org
0 !LDRAW_ORG Configuration UPDATE 2009-09-05
0 !LICENSE Redistributable under CCAL version 2.0 : see
```

CAreadme.txt

0 !TRANSLATION COLOURS de

"Black" = "Schwarz"

"Blue" = "Blau"

"Green" = "Grün"

"Dark_Turquoise" = "Dunkeltürkis"

"Red" = "Rot"

"Dark_Pink" = "Dunkelrosa"

and here a sample for the parts.txt:

(Please note that the preceeding flags '~' (tilde) and '_' (underscore) are forced to be carried on in the translation.)

0 LDraw.org Configuration File

0 Name: parts.txt

0 Author: LDraw.org

0 !LDRAW_ORG Configuration UPDATE 2009-09-05

0 !LICENSE Redistributable under CCAL version 2.0 : see

CAreadme.txt

0 !TRANSLATION PARTS de-DE

"Animal Bat Wing 9 x 9 with Axle" = "Tier Fledermaus Flügel 9 x 9 mit Achse"

"Animal Bird" = "Tier Vogel"

"Animal Bird with Parrot Pattern" = "Tier Vogel mit Papagei Dekor"

"Animal Cat Crouching" = "Tier Katze kauern"

"~Technic Pneumatic Cylinder with 2 Inlets Medium Slide" =

"~Technik Pneumatik Zylinder mit 2 Einlässen Mittel Gleiter"

"~Moved to 75974" = "~Verschoben nach 75974"

"_Brick 2 x 2 White" = "_Stein 2 x 2 Weiss"

"Brick 1 x 1 with Blue \"0\" Pattern" = "Stein 1 x 1 mit Blau \"0\" Dekor"

and here a sample for the categories.txt:

0 LDraw.org Configuration File

0 Name: categories.txt

0 Author: LDraw.org

0 !LDRAW_ORG Configuration UPDATE 2009-09-05

0 !LICENSE Redistributable under CCAL version 2.0 : see

CAreadme.txt

0 !TRANSLATION CATEGORIES de-DE

"Animal" = "Tier"

"Antenna" = "Antenne"

"Arch" = "Bogen"

and here a sample for the keywords.txt:

0 LDraw.org Configuration File

0 Name: keywords.txt

0 Author: LDraw.org

0 !LDRAW_ORG Configuration UPDATE 2009-09-05

0 !LICENSE Redistributable under CCAL version 2.0 : see

CAreadme.txt

0 !TRANSLATION KEYWORDS de-DE

"cone" = "Konus"

"exhaust" = "Auspuff"

It is not forced for the translations to avoid double entry part description <-> keywords as it is difficult to compare those entries but should be avoided to minimize used discspace.

Applications should use translations only for presenting a string to the user and not to decide anything based on that translation string.

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Documentation: !AVATAR Language Extension

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Maintained by: LDraw.org Standards Board

Revision: 1, 2024-06-19

Revision History:

1: Initial

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Purpose

Outlines the !AVATAR Meta statement that provides information to display a typical example of a part in a given category.

Scope

The !AVATAR meta command is an optional command intended to allow an editor to display a typical example of a part in a given category (as defined by the !CATEGORY meta). Defined !AVATAR statements will be located within the LDConfig.ldr file. For an editor that chooses to implement this specification, if an !AVATAR statement for a given !CATEGORY is not defined, the choice of avatar is at the option of the editor developer.

!AVATAR Syntax

```
0 !AVATAR CATEGORY "<category>" DESCRIPTION "  
<description>" PART <rotation matrix> "<part file>"
```

<category> is the category name from the list of approved !CATEGORY categories

<description> is a short (approximately 4 words or less) description of the category. This will typically be the pluralized form of the category

<rotation matrix> is the 9 element rotation matrix as defined by the LDraw File Format Type 1 line definition. Only rotations are allowed in this matrix, not scale factors or skews.

<part file> is the part file reference as defined by the LDraw File Format Type 1 line definition

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Maintained by: LDraw.org Library Administrator

Revision: 2.4 (03-Feb-2023)

Revision History:

- 1 - Original
- 2 - Reorganized and consolidated other sources into this document
- 2.1 - Clarified requirements for TEXMAP geometry and fallback
- 2.2 - Added NOCLIP and !KEYWORDS requirements for stickers
- 2.3 - Clarified description requirements for patterns on mold variants parts

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Introduction

This document describes additional specifications for parts to be included in the LDraw.org Official Parts Library (Official Library). Unless specifically contradicted by this document all requirements within the [LDraw File Format 1.0.2 specification](#) also apply to part files within the Official Library.

These rules have evolved over time, and there may be files in the Official Library that do not conform to these rules. Legacy parts should not be edited solely for the purpose of making them conform. However, if the part is changed and submitted to the Parts Tracker for any other reason, non-conformant lines must be updated.

General File Naming

File Name Limitations

Filenames must not be longer than 25 characters (including the extension) and must only contain the following characters:

a-z, 0-9, _, -

(To avoid confusion with English grammar, comma is not a permitted character.)

Note: The file format specification states that filenames are case-insensitive. However, all official parts will be issued with lower-case only names.

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- [Back Face Culling \(BFC\)](#)
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Extension

All part files in the LDraw Parts Library are required to carry the `.dat` extension.

Part Numbering

Part numbers for official parts must conform to the [Official Library Part Number Specification](#)

Texture Image Files

See [Official Library Part Number Specification](#) for texture image file name rules.

The encoding format of texture images is specified as PNG. Therefore texture images will have an extension of `.png`

General File Content

Numbers

Precision

In general, three decimal places are sufficient for parts, subparts, and primitives representing portions of parts that are not intended to be scaled (for example, studs, pegs, peg-holes, clips, hinge ends, etc.). Four decimal places should be used for high-res primitives and any primitives designed to be scaled (for example, cylinder sections, boxes, rectangles, discs, edges, etc.) as this allows such primitives to be scaled by a factor of ten while still preserving three decimal places of accuracy. The [primitives reference](#) indicates which primitive families are not supposed to be scaled.

Format

Trailing zeros must be removed (i.e. 1.5, not 1.500). Leading zeros unless immediately before the decimal point must be removed (i.e. 1.5, not 01.5). The leading zero before the decimal point is optional (i.e. 0.5 and .5 are both valid).

Geometry

Matrix restrictions

The matrix specified for sub-files (line type 1) must not be singular nor contain columns or rows of all zeros.

Co-planar/Co-linear Tolerance and Concave Quads

Quads and triangles with co-linear vertices, quads with non-co-planer vertices, and concave quads are disallowed per the LDraw File Format. However, the tolerances for this co-linearity, co-planarity are not defined.

To prevent co-linear vertices in quads/triangles and concave quads, every interior angle in a quad or triangle must not be greater than 179.9 degrees or smaller than 0.025 degrees. Concave quads must be split into two triangles.

A quad is considered co-planar if it is split into two triangles and both triangles are co-planar with each other. There are two ways to split a quad

into two triangles, and both must result in co-planar triangles. The triangles are considered co-planar if the angle between their surface normals is less than or equal to 3 degrees. However, an angle of less than 1 degree is required unless justification is given by the part author. This is especially true for large quads where the "warp" can be seen at normal viewing magnifications.

Overlapping and Intersecting Line Types

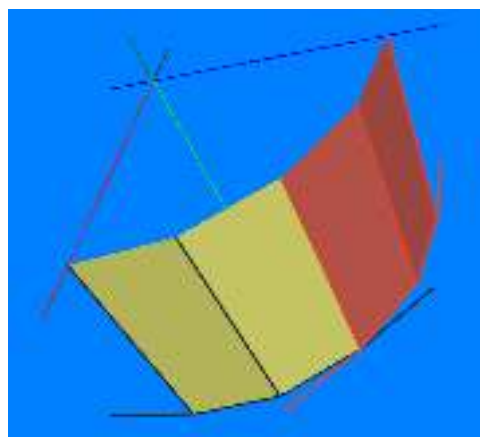
All or part of a line (line type 2) may not overlap all or part of any other line (line type 2).

All or part of a conditional line (line type 5) may not overlap all or part of any other conditional line (line type 5), with the exception of complementary conditional lines for curved primitives. Complementary conditional lines are placed at the end of curved primitives where those curved primitives may join with either a continuation of the curve, or with other geometry. In this case, the control points that are off of the edge of the curved primitive are placed so they are tangent to the curve.

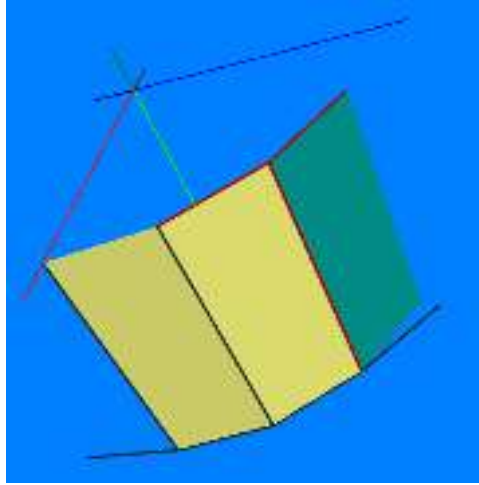
If two such curves are placed next to each other (for example to turn a quarter cylinder into a half cylinder), the conditional lines on the edge of each primitive will overlap exactly, and their control points will complement each other such that at most one will be rendered at any given time, and one of the two will always be rendered when appropriate. When other geometry is placed against such a curved primitive, nothing needs to be done if the geometry is tangent to the curve. If it is sloped further, then a complementary conditional line will be placed on the edge of that geometry, with the control points past the edge of the geometry configured to be tangent to the curve, and the two end points of the conditional placed to exactly overlap the complementary conditional line on the edge of the curved primitive.

Example:

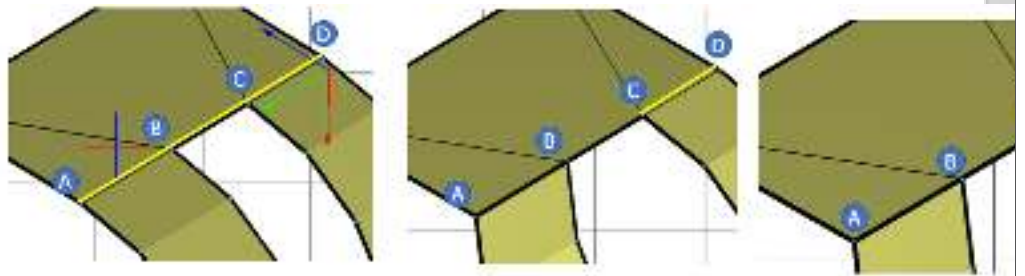
The first and last conditional lines in a cyli primitive have a control point that is tangent to the cylinder. When two such cyli primitives join, the two conditional lines complement each other, like so:



This shows the extra complementary conditional line an author has to add if a surface is not placed in the tangential plane.



Part (but not all) of a conditional line (line type 5) may overlap all (but not part) of a line (line type 2). Consider the line A-B-C-D



- If AB and CD need to be conditional lines and BC a line, it is acceptable to define this as a line BC and a single conditional line AD
- If AC needs to be a line and just CD needs to be a conditional line, it is **not** acceptable to define the conditional line as either AD or BD
- If AD is a line, there are no valid conditional lines along the length of AD

Lines (line type 2 or 5) may intersect any other line (line type 2 or 5).

Every effort must be made to remove overlapping co-planar polygons (line type 3 or 4). Where overlapping polygons are unavoidable, the overlap must be kept to the absolute minimum.

Every effort must be made to remove intersecting (non co-planar) polygons (line type 3 or 4). Where intersecting polygons are unavoidable, the intersection(s) must be kept to the absolute minimum.

Note: Primitives that overlap/intersect other primitives or polygons (line type 3 or 4), provided that the overlap is small, are considered acceptable. The intention is **not** to force part authors to in-line primitives. However, there may be some cases where it is better to use (say) a 3-8cyli and a quad than a 1-2cyli; case-by-case decisions are left to the parts review process.

Orientation/Origin

In general, parts that have studs should be oriented such that the top studs point up (-y) and the bottom tubes point down (+y).

For parts with studs, the origin of the parts will be centered on the top most stud group. The bottom of studs should lie on the x-z plane (i.e. the bottom of the studs should be at y=0)

For hinge or hinge like parts, the origin should be at the rotation point

These rules are generic. Look at other similar parts if it is unclear and if uncertainty remains, ask. If some other orientation/origin makes sense, the

reasoning should be made when the part is submitted to the Parts Tracker.

Stud Orientation

The individual studs on a part should be rotated such that the stud logo appears as it would on the real part. Since the generic stud.dat does not show a logo, rotation should be checked by using a LDraw program designed to show logo.

Subparts that use studs should not be mirrored as this would cause the logo to appear mirrored. In this case, the studs can be moved from the subfile to the parent file which will avoid mirroring but still allow reuse of symmetrical geometry.

Duplicate Line Types

Identical line type 1-5 lines are not permitted.

The ordering of the vertex points in lines, quads, triangles, and conditional lines (line types 2-5) are not considered when checking for identical lines. The control points are not considered when checking for identical conditional lines (line type 5).

Colour restrictions

Polygons (line types 3 and 4) shall not use colour 24.

Lines (line types 2 and 5) must use colour 24.

All colour numbers used in part files **must** be present in LDConfig.ldr, with the exception of pattern/sticker geometry, which may use Direct Colours (see [LDraw File Format Spec](#)).

LDConfig.ldr will only contain colours that have at some point in time been used by LEGO as the non-painted colour of materials used in LEGO-branded toys, plus ldraw.exe's special colour codes 16 and 24. This includes materials such as the metal track for 9V train tracks, but does not include the colours of paint used to decorate parts or ink used on stickers.

In general, metallic colours on stickers, patterned parts, and other printed materials should use the LDraw Metallic Colours and not Chrome or Pearl colours. There are exceptions that will be handled on a case by case basis.

META Commands

Header Meta Commands

Official parts must conform to the [Official Library Header Specification](#).

Body Meta Commands

Only the following meta commands are permitted in the body of official parts:

- 0 // style comments
- All meta commands specified in [Language Extension for Back Face Culling \(BFC\)](#)
- All meta commands specified in [Texture Mapping \(!TEXMAP\) Language Extension](#)

Back Face Culling (BFC)

All official parts will contain 0 BFC CERTIFY CCW and have valid quad/triangle winding to support this. See [Language Extension for Back Face Culling \(BFC\)](#) for reference.

!TEXMAP

Sufficient !TEXMAP <geometry2> and/or <geometry3> lines (as outlined in the [Texture Mapping \(!TEXMAP\) Language Extension](#)) must be included such that the part renders correctly in non-!TEXMAP supporting programs. It is highly encouraged that the included geometry reflect the pattern created by the !TEXMAP (even if it is reduced in resolution or colors) but this is not required.

Generated File Meta Commands

Meta commands used by programs to generate file geometry may be left in the part file provided they are correctly commented out. It is desirable to add a comment to mark the end of the generated section.

Shortcut Part Regulations

Shortcut Part Numbering

See [Official Library Part Number Specification](#) for shortcut part number rules.

General Rules for Shortcuts

Shortcut files are used when several separate parts are typically delivered from LEGO pre-assembled or are frequently used in combination.

Shortcuts are typically made of the unpatterned (or plain) version of an assembly. Shortcuts of patterned assemblies will only be considered on a case by case basis. Assemblies that do not conform to this rule can remain in the library and not be made obsolete.

When parts are used in assemblies (provided as such by LEGO and impossible to separate in normal use) their description begins with ~ so they don't appear in part selection lists. LDraw users should use the assembly shortcuts instead.

To further clarify:

- For assemblies without moving parts, it's simple: all parts composing the assembly are ~ and an assembly shortcut is provided. Several shortcuts may be needed to account for different colour combinations.
Example:
NXT light sensor 55969 shortcut, composed of ~ parts 54732, 55972, 55816, 55970, 55966.
- For assemblies with moving parts, a shortcut should be provided for each mechanical entity. This way the LDraw user can compose their own assembly in any position. An overall assembly is provided as well, to simplify user life when default position is usable, and for bill of materials usage. Different positional forms of the same combination of parts should use an additional -fN suffix to distinguish them.
Example:
NXT motor has a case assembly 54715c01 and an axle hub assembly 54725c01. Case assembly 54715c01 is made of ~parts 54714, 54715, 54716, 54717 and 54732. Hub assembly 54725c01 is made of two

54725 ~parts. Complete motor assembly 53787 is made of 54715c01 + 54725c01.

- In case one of the moving assemblies is reduced to a single part (eg. axle hub of PF motors), the description of this part is not tilded, so that the user can pick it easily to create his own position combination. For example, the small linear actuator has 3 sections moving independently:
 - the body, assembly 92693c01, composed of ~parts 92693 and 92697.
 - the rotating axle, single part not tilded 92695.
 - the piston, single part not tilded 92696.

Two complete assemblies are provided, contracted, and extended (92693c01-f1 and 92693c01-f2).

Flexible Parts

Flexible parts are parts that are made out of a non-rigid material and are intended by LEGO to deform. Examples include hoses, ropes, strings, springs, fabrics, rubber bands, and plastic sheets

These parts are typically modelled as a series of discrete part pieces that are then "formed" to simulate curvature. This places them in the shortcut category, and their descriptions will follow the rules outlined for shortcuts (i.e. use of the "~" character).

Flexible parts are always modelled in their flat or at rest state as the primary version. Subsequent formed versions shall have a suffix added to the part number as specified in the [Official Library Part Number Specification](#).

Sticker Parts Regulations

Sticker Numbering

See [Official Library Part Number Specification](#) for sticker number rules

Sticker Description

The description of the part should begin with 'Sticker ...' followed by the dimensions if rectangular or round, or a description of the shape, and then a description of the sticker. The description should not end with 'Pattern'.

Dimensions should be specified in studs (where 1 stud = 20 LDraw units) and as [z-dimension] x [x-dimension]. Additional rules regarding dimension numbers are contained in the [Official Library Header Specification](#)

Examples:

Sticker 2.6 x 5.5 ...

Sticker 3.1 x 3.1 Round ...

Sticker Minifig Torso ...

Sticker Minifig Shield Triangular ...

Sticker !KEYWORDS Entry

All stickers and sticker shortcuts should, at a minimum, include at least one !KEYWORDS entry in the form of Set <set number> where <set number> is the number of an official LEGO set in which that pattern appears. Only one set need be listed but more can be if desired.

Sticker Geometry

Stickers should be created flat. If the sticker is designed to be placed on the curved surface of a part or folded around the edge of a part, a 'formed' version of the sticker part may be modelled in addition to the flat version.

The sticker should be modelled as a 0.25 LDU thin box (or whatever shape sticker is in), without any edge lines (type 2 lines), and coloured with main colour 16. The sticker should be oriented such that the top face is at -0.25 Y and parallel with the X-Z plane. The pattern with the hardcoded colours is on the top face.

The origin of a formed sticker part may match that of the part for which it is intended or be noted in a !HELP header line.

The sticker pattern is modelled in its true colours; they are not modifiable from the outside. All colours of the top surface must be matched. The back side and sides of the sticker must use colour 16. Print on a transparent plastic sticker sheet should use colour 16 on the transparent part of the top surface.

For stickers printed on a transparent medium, the type 1, 3, and 4 lines that comprise the pattern shall be preceded by a 0 BFC NOCLIP statement. If there is non-pattern geometry after the pattern geometry, the pattern geometry shall be ended by a 0 BFC CLIP statement.

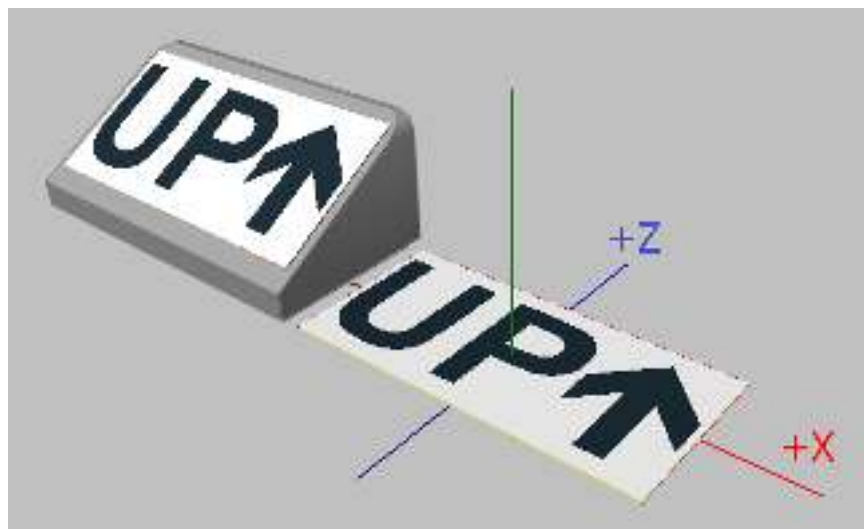
Conditional lines (type 5 lines), in colour 24, should be provided around curved edges and on the curved surface of formed sticker parts. Solid edges (type 2 lines) are also allowed for folds in formed sticker parts, but not around the edges of the sticker.

Sticker Orientation

Sticker patterns should be oriented so that top is aligned with +Z axis.

Top is defined as either the logical top of a picture, text, etc. or, in the case of more abstract patterns, what the top is when the sticker is applied.

When in doubt, sticker should be oriented such that X value is greater than Z (Landscape).



Sticker Backing Boxes

The backing box is defined as the sides and bottom surface of the sticker

See the [LDraw.org Library Sticker Box Standard](https://www.ldraw.org/library/sticker-box-standard)

Sticker Shortcuts

Shortcuts that consist of one part (which may be a composite part) and one or more stickers are allowed in the official library.

See [Official Library Part Number Specification](#) for sticker shortcut part number rules.

Additionally:

- The shortcut may not include more than one part, although that part may be a shortcut part.
- The sticker(s) must be applied in the same location(s) as shown in the instructions for the official LEGO model from which the sticker(s) originate(s).

Patterned Part Regulations

Patterned Part Numbering

See [Official Library Part Number Specification](#) for patterned part number rules.

Pattern Part General Rules

Patterned parts are considered both parts with printing on the plastic and parts that are moulded with multiple plastic colors

Some patterned parts have never been issued as plain, unpatterned parts by LEGO. Since this printing is easily removed, and LDraw parts can come in any variety, all patterned parts should be based on another plain, unpatterned part in the library. This means that a plain version of the patterned part should be created even if a plain version has never been produced by LEGO.

Pattern Description

The description of a patterned part should in the following format:
`<base part description> with <pattern description> Pattern`

Where:

- `<base part description>` is the description of the non-patterned version of the part. Note: For parts with mould variants (e.g. tiles with or without groove), the additional qualifier for the mould is only required if the same pattern is printed on more than one variation. Example: "Minifig Head with Solid Stud with ... Pattern" can be shortened to "Minifig Head with ... Pattern" if the patterned being modeled only appears on the "Solid Stud" variation.
- `<pattern description>` is a reasonably detailed description of the pattern. If unsure, use of the description on other inventory websites is encouraged.

Pattern !KEYWORDS Entry

A patterned part should, at a minimum, include at least one !KEYWORDS entry in the form of `Set <set number>` where `<set number>` is the number of an official LEGO set in which that pattern appears. Only one set need be listed but more can be if desired.

It is encouraged, but not required, that a cross reference number from an external inventory website (e.g. Bricklink, Rebrickable, BrickSet, etc.) be included in the !KEYWORDS. In this case the entry should be <website name> <part number on website> where <website name> is the name of the website and <part number on website> is that specific website's number for the pattern part (example: Bricklink 973pb0042).

Example:

```
0 !KEYWORDS Set 10442, Bricklink 973pb0042
```

Errors in either one of these !KEYWORDS entries should not be cause for a hold vote on the Parts Tracker since this is simply data to make locating specific patterns easier.

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Documentation: Official Library Part Number Specification

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Maintained by: LDraw.org Library Administrator

Revision: 2 (8-May-2024)

Revision History:

Revision 1: Original Revision 2: Clarified p/c/d suffix codes.

This is an ratified, official LDraw.org document. Non-administrative changes can only be made with the approval of the maintainer.

Introduction

This document outlines the LDraw part numbering schema for the Official Library. While the goal is to be comprehensive, it may not outline all situations involving the numbering of parts for the Official Library. As these rules have evolved over time, there may be part in the Official Library that do not conform. This legacy numbering remains in place for backwards compatibility reasons. Do not submit parts to the Parts Tracker if you are unsure. When in doubt, please ask! You can post to everyone on the [LDraw Forums](#), or email the [parts library administrator](#).

General Part Numbering

Known Part Numbers

If the part number (Design ID) used by LEGO is known, then that number should be used.

Where to Find Part Numbers

For modern parts, these numbers are often moulded in a discreet place on the part. These are typically four digit numbers in the range 2001-9999, or five-digit numbers. Don't be confused by a number of the form $n-nn$ (eg. 4-01), or even single numbers or letters on older parts - these are probably something to do with identifying individual moulds, they are not the part number.

Part Numbers vs. Element IDs

- [General Patterned Part Numbering](#)
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The numbers listed next to parts at the back of instruction manuals are not the part numbers (also called Design IDs) LDraw.org uses as its nomenclature, but the Element IDs or Item IDs, a number that signifies both the part shape and color. Since LDraw parts can be any color, LDraw does not use Element IDs but solely Design IDs. For example, a Brick 1x2's Design ID is "3004" while a Tan Brick 1x2's Element ID is "4109995". Hence you will not find 4109995 in the LDraw parts library.

Unknown Part Numbers

If LEGO part number is not known, "u" prefix is used (e.g. u1234.dat). The part in question gets a file name with uNNNN.dat format, these filenames thus ranging from u1.dat to u9999.dat. This number will be requested from the Parts Library Admin prior to submitting the part to the Parts Tracker. The part will retain the u-prefixed number as long as the part's true number is unknown. If the LEGO part number is identified during the review process, then the part will renumbered and the u-prefixed number will become available for re-use.

Prior to 2010, a three-digit part number was used. That range of numbers, however, was exhausted. During development, and for posting to the Parts Tracker, the three-digit number was preceded with an "x" (e.g. x123.dat), and upon inclusion in an official release, the "x" was removed. Parts that were on the Parts Tracker in February 2010 and not listed by Peeron, do not need be renamed to u9NNN numbers. They will continue to be renamed xNNN.dat -> NNN.dat upon release.

For backwards compatability reasons, nonconforming parts already in the Official Library will not be renumbered solely for the purpose of conforming to this rule.

Mould Variants/Different Versions or Parts with the Same Number

If the parts have different numbers, the different LEGO (design ID) numbers will be used. More commonly, the different variations will have the same number. This is managed by adding a single alpha character to the end of the base part number (e.g. 3626a.DAT and 3626a.DAT, the solid stud and hollow stud versions of the minifig head). If, prior to the second version being authored, one version of the part was released using just the official part number (no alpha-character suffix), it will be renumbered/moved to use an alpha-suffix. If the official part number is unknown, then separate uNNNN numbers will be used for each version.

Occasionally a part is included with geometry that is not quite correct or the part number itself is incorrect. In this instance, in order to preserve backward compatability, when a corrected part is made, it's part number will follow these rules and the non-alpha suffix part will be made obsolete.

Parts on Sprues or Runners

Some small parts are delivered from LEGO still connected to a sprue (or runner) containing several different parts (e.g. the castle feathers or the coins). These are numbered in the same way as multi-version parts - by adding a single character suffix to the end of the part number.

Subparts

Subparts, which are stored in the parts\s subdirectory, should bear some relationship to their "parent" main part. Where there is extensive shared geometry, subparts may be used by parent parts with a different main number. They are numbered by appending a sNN suffix to the part number of the main part.

Examples:

s\1234s01.dat for 1234.dat

s\1234p56s01.dat for 1234p56.dat>

Primitives

Primitives are essentially highly reusable components, and for ease of use have highly coded names. Existing primitives are described in the [Primitives Reference](#). If there is a need for a new class of primitives, the [Parts Library Admins](#) should be contacted prior to submission.

Stud groups

Stud groups are provided to reduce the size of part files with many regularly spaced studs. The naming convention is as following:

stugN-XxZ.dat

Where:

- **N** = type of stud (regular stud, hollow stud, underside stud, ...)
- **X** = number of studs on the x axis
- **Z** = number of studs on the z axis

To prevent an overload of the library with all sort of combinations, the numbers on the x and z axis are limited to:

- 1xZ stugs
- Xx1 stugs (Due to stud orientation you cannot rotate a 1xZ stud group by 90 degrees to get a Xx1 stud group)
- XxZ stugs where X==Z

Texture Image Files

Texture image files will have the same number as its parent file. If more than one texture image is used for a part, a suffix of the part author's choice is added to the files. Examples:

4141698a.png for 4141698a.dat

6092p01wall1.png and 6092p01wall2.png for 6092p01.dat

Third Party Parts

Third party parts will have a tNNNN.dat format where NNNN is a number assigned by the Library Administrator

Shortcut (Assembly) Part Numbering

General Shortcut Part Numbering

Shortcut files used when several separate parts are typically delivered from LEGO pre-assembled or are frequently used in combination. These are numbered using the official LEGO number for the composite part, if known, or by appending cNN to the part number of one of the individual parts.

Flexible parts such as springs, hoses, ropes, chains, or cloth can exist in more than one configuration.

If the flexible part is of a nature that smaller component parts are used to mimic the curving of the part (e.g. ribbed hoses, flexible axles), these parts with have a suffix of kNN

In all cases, the final assembly will have a suffix of –fN. The default or "at-rest" position will always be –f1

Patterned Part Numbering

General Patterned Part Numbering

In LDraw, a part with printing is called patterned. The decorative printing is the pattern. LDraw has it's own pattern numbering scheme, do not use the pattern code listed on other inventory websites.

The number for a patterned part is the same as the number for plain, or base, part, plus a 'p' and a 2 (or 3) digit code:

BBBBBpCC(C).dat

Where:

BBBBB is the 3-5 digit base number

CC(C) or XXXX is the code for the pattern. The CC(C) code can have a mix of letters and numbers. The XXXX code is only numbers.

Patterned Part Code

Typically, the first digit of the code helps to identify which theme or product line the pattern appears in (themes listed below). This is not a requirement, both because many patterned parts appear in multiple themes, and because this approach to pattern-numbering has evolved over time. If a theme is full (i.e. 0-z are in use), then any available code can be used. If all codes are in use (i.e. 00-zz) for a given part number then the first available XXXX number should be assigned (e.g. 0001)

Range	Assignment
0X	General/Miscellaneous and Town
1X,2X	Town, including Paradisa
3X	Pirates, Soldiers, Islanders
4X	Castle
5X,6X	Space
7X,8X,9X	Modern Town
aX	Action (Adventurers, Aquazone, Alpha Team, Rock Raiders)
bX	Superheroes

Range	Assignment
cX	Control Panels, dials, gauges, keyboards, readouts, etc. / Superheroes (minifig parts)
cXY	Patterned parts for Collectable Minifigures (heads, torsos, arms, hips, legs) from numbered series: - Note: X cannot be 0 - X=series number : 1,2,3,...a,b,c,..., Y=sequence : 1-9,a-h (a=10...h=17) - X=z for Collectable Minifigures from accessory packs (where these differ from series versions)
dX	Studios
dXY	Patterned parts for Collectable Minifigures (heads, torsos, arms, hips, legs) from unnumbered series: - Note: X cannot be 0 - X=1 for Simpsons series 1; Y=1-9,a-g - X=2 for The LEGO Movie; Y=1-9,a-g - X=3 for Simpsons series 2; Y=1-9,a-g - X=4 for Disney; Y=1-9,a-i - X=5 for 2016 German Football Team; Y=1-9,a-g - X=6 for The LEGO Batman Movie; Y=1-9,a-k - X=7 for The LEGO Ninjago Movie; Y=1-9,a-k - X=8 for The LEGO Batman Movie series 2; Y=1-9,a-k - X=9 for Wizarding World; Y=1-9, a-m - X=a for The LEGO Movie 2; Y=1-9, a-k - X=b for Disney series 2; Y=1-9, a-i - X=c for DC Super Heroes; Y=1-9,a-g - X=d for Harry Potter series 2; Y=1-9,a-g - X=e for Looney Tunes; Y=1-9,a-c - X=f for Marvel Studios; Y=1-9,a-c - X=g for The Muppets; Y=1-9,a-c - X=h for Disney 100 Years; Y=1-9,a-i - X=i for Marvel Studios series 2; Y=1-9,a-c - X=j for 2012 Team GB; Y=1-9
eX	Nexo Knights
fX	Fabuland and Scala or Castle (minifig parts)
gX	Soccer, Basketball
hX	Harry Potter
jX	Indiana Jones
kX	Cars (Disney Pixar)
mX	Middle Earth (Lord of the Rings), Elves

Range	Assignment
nX	Ninja
qX	Pharaoh's Quest
rX,sX	Star Wars
tX	General Textual Patterns (lettering and numbers) and Trademark items (Corporate Logos, etc)
uX	Extended textual patterns or Modern Town/City (minifig parts)
vX	Extended textual patterns
wX	Extended textual patterns or Western (minifig parts)
xX	Miscellaneous Licenses (SpongeBob SquarePants, Ideas)
yX	Racing (Racers, Tiny Turbos, Speed Champions)
zX	Brickheadz

Sticker Part Numbering

General Sticker Part Numbering

Official LEGO numbering for sticker sheets has changed over time

1. no copyright date sheets - 4-digit numbers: 3xxx, 4xxx or 6-digit numbers: 004xxx
2. no copyright date sheets - 6-digit numbers: 168xxx, 169xxx, 197xxx, 820xxx, 821xxx
3. ©1993 / 1994 - 6-digit numbers: 168xxx, 169xxx, 822xxx; a few 7-digit numbers: 4100xxx
4. ©1995 - 2000 - combo 5-digit/7-digit numbers: 71xxx/41xxxxx, 72xxx/41xxxxx
5. ©2000 - 2003 - combo 5-digit/7-digit numbers: 2xxxx/41xxxxx, 4xxxx/41xxxxx, 4xxxx/42xxxxx; a few 7-digit numbers: 42xxxxx
6. ©2004 onwards - combo 5-digit/7-digit numbers: 5xxxx/4xxxxxx, 6xxxx/4xxxxxx, 7xxxx/4xxxxxx, 8xxxx/4xxxxxx
7. ©2012 onwards - combo 5-digit/7-digit numbers: 1xxxx/60xxxxx

Sticker parts should be based on the sticker sheet number, if known, with a character suffix (a, b, c, etc.) for each sticker. If the sticker sheet contains only a single sticker or multiple copies of the same sticker, a suffix (a) should still be used. If the stickers are numbered on the sticker sheet, the printed sticker number and the character suffix should correspond to the sequence where a=1, b=2, ... , z=26, aa=27, ab=28, etc. Four digit numbers should be prefixed with two zeros to avoid clashes with LEGO parts (e.g. 004845a.dat, not 4845a.dat). Stickers from sheets with combo 5-digit/7-digit numbers should use the 7-digit number.

Sticker Sheets with No Number

Stickers from sheets with no printed number or of unknown number should use the next available **sN**, **sNN** or **sNNN** number.

Formed Stickers Parts

Formed versions of stickers to match the surface of a part should use the flat sticker part number with a **cNN** or **cXXXX** suffix. The **NN** suffix can have a mix of letters and numbers. The **XXXX** suffix is only numbers. If all **NN** suffixes are in use (i.e. 00-zz) for a given part number then the first available **XXXX** number should be assigned (e.g. 0001)

Sticker Shortcuts

Sticker shortcuts should use the number of the part that the sticker is applied to with a **dNN** or **dXXXX** suffix. The **NN** suffix can have a mix of letters and numbers. The **XXXX** suffix is only numbers. If all **NN** suffixes are in use (i.e. 00-zz) for a given part number then the first available **XXXX** number should be assigned (e.g. 0001)

Part Numbering Quick Reference Table

Part number pattern (<i>a</i> = alpha, <i>N</i> = numeric, <i>x</i> = alphanumeric)	Usage
<i>NNN</i>	Regular part of unknown LEGO number (see also <i>uNNNN</i> , below).
<i>NNNN</i> or <i>NNNNN</i>	Regular part of known LEGO number.
<i>NNNcNN/XXXX</i> , <i>NNNNcNN/XXXX</i> , <i>NNNNNcNN/XXXX</i>	Shortcut assembly of part <i>NNN</i> , <i>NNNN</i> or <i>NNNNN</i> with other parts or formed version of flexible part <i>NNN</i> , <i>NNNN</i> or <i>NNNNN</i> .
<i>NNNcNN-fn</i> , <i>NNNNcNN-fn</i> , <i>NNNNNcNN-fn</i>	Positional variant of shortcut assembly of movable parts, comprising part <i>NNN</i> , <i>NNNN</i> or <i>NNNNN</i> with other parts.
<i>NNNdNN/XXXX</i> , <i>NNNNdNN/XXXX</i> , <i>NNNNNdNN/XXXX</i>	Shortcut assembly of part <i>NNN</i> , <i>NNNN</i> or <i>NNNNN</i> with a sticker as used in an official set.
<i>NNNpCC(C)/XXXX</i> , <i>NNNNpCC(C)/XXXX</i> , <i>NNNNNpCC(C)/XXXX</i>	Patterned version of regular part.
<i>NNN[a-z]pCC(C)/XXXX</i> , <i>NNNN[a-z]pCC(C)/XXXX</i> , <i>NNNNN[a-z]pCC(C)/XXXX</i>	Patterned version of regular part variant.

Part number pattern (<i>a</i> = alpha, <i>N</i> = numeric, <i>x</i> = alphanumeric)	Usage
<i>s/NNNsNN</i> , <i>s/NNNNsNN</i> , <i>s/NNNNNsNN</i>	Subfiles - stored in the parts/s directory.
<i>NNNa</i> , <i>NNNNa</i> , <i>NNNNNa</i>	Regular part variant (<i>a</i> = a, b, c for successive versions of part, sequenced chronologically), or mini-part (<i>a</i> = a, b, c for each mini-part on the same runner or included in a small bag), or to maintain backward compatibility if an incorrect number is released into the official library.
<i>sNNNN</i>	Sticker of unknown LEGO number.
<i>tNNNN</i>	LEGO-compatible third-party part.
<i>uNNNN</i>	Part of unknown LEGO number

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